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When regional policies fail: An evaluation of Indonesia's Integrated Economic Development Zones☆

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ABSTRACT

We study Indonesia's Integrated Economic Development Zone (KAPET) program. Although firms in KAPET districts paid lower capital taxes, the program's incentives neither stimulated entry nor increased output, and KAPET districts experienced similar development outcomes relative to non-treated districts. To investigate whether regional policies could be more optimally redesigned, we develop a quantitative spatial model where place-based capital tax cuts affect multiple sectors and impact a transfer system to finance local public goods. We find that capital incentives in KAPET areas would have been more welfare and growth enhancing if they had been accompanied by additional place-based improvements.

1. Introduction

In developing countries, manufacturing firms are often characterized by small scale and persistently low growth. Such firms are constrained by restricted access to markets, capital, and technologies, as well as by unfavorable investment climates (Tybout, 2000). To spur industrial growth, many countries have established special economic zones, a particular place-based policy that provides tax incentives or reduced regulatory burdens for firms locating in specific places. Such policies are common in the developing world; examples include China's special economic zones (Wang, 2013), export processing zones in Bangladesh, Thailand, and Vietnam, and free trade zones in Honduras (Farole and Akinci, 2011). While some countries choose to establish these zones in favorable areas with good market access, others use them to attract investment to poorer regions.

If threshold effects and multiple equilibria are important features of economic growth, these policies may attract enough manufacturing activity to constitute a "big push" that generates large increases in productivity, improving welfare not only in the affected localities but also throughout the country (e.g., Rosenstein-Rodan, 1943; Murphy et al., 1989; Azariadis and Stachurski, 2005). Although place-based tax cuts may be initially costly, they may also generate productivity spillovers for existing firms and encourage entry. This could create a self-sustaining, virtuous circle of development, with growth dividends that pay for the initial tax cuts.

However, the relative costs and benefits of place-based policies are difficult to discern. If workers are imperfectly mobile, place-based policies could encourage more productive reallocations of labor, but they may also just reshuffle economic activity from one place to another without increasing aggregate welfare (Bartik, 1991; Glaeser and Gottlieb, 2009; Kline and Moretti, 2014). Tax cuts could also make it harder for lower to middle-income countries to provide for public goods.

This paper investigates the effects of a large place-based intervention in Indonesia known as the Integrated Economic Development Zone (*Kawasan Pengembangan Ekonomi Terpadu*, or KAPET) program. This program, which began in the late 1990s, provided substantial capital tax-breaks for firms in certain districts in the Outer Islands of Indonesia, a country with large regional differences in per-capita income, a sizable informal sector, and a history of policies to promote inclusive growth. To study this policy, we combine treatment variation with high quality

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panel data on demographic outcomes, measures of regional growth, poverty, employment, informality, and manufacturing outcomes. Using a series of reduced form exercises, we estimate the average treatment effect of the KAPET program on treated districts for this rich set of economic outcomes.

When estimating the impact of place-based policies like the KAPET program, a central concern is that omitted variables influence selection into the program and also affect outcomes. Because Indonesia's Outer Islands tend to be less developed than the Inner Islands (Java and Bali), we document that on average, program districts were poorer, further from Jakarta, and less populated than non-treated districts. This endogenous program placement creates a negative targeting bias, meaning that naive treated vs. non-treated comparisons may underestimate program impacts.

We adopt several approaches to improve identification. First, we restrict the non-treated comparison group to include only districts in the Outer Islands, which are more comparable to KAPET districts. Next, we combine a difference-in-differences design with an inverse probability weighting (IPW) approach that explicitly adjusts for potential *ex ante* differences between KAPET and non-KAPET districts. This approach uses propensity-score reweighting to adjust the contribution of non-treated district trends to the counterfactual in accordance with their odds of treatment. After implementing this IPW approach, we verify that pre-trend differences between KAPET and comparison districts disappear. As an alternative, we also use the synthetic difference-in-differences approach of Arkhangelsky et al. (2021), which combines attractive features of traditional difference-in-differences designs with the method of synthetic controls.

We find that along many dimensions, KAPET districts experienced no better development outcomes than their non-treated counterparts. We find no significant impacts on migration, population growth, employment outcomes, or consumption expenditures. We also find that the program was not associated with increased economic growth or nighttime light intensity, another commonly used measure of regional output (Henderson et al., 2012; Olivia and Gibson, 2013). To verify that our null estimates of the KAPET program are precise, we use a simulated power analysis to calculate the minimum detectable effect sizes that our estimators can reliably estimate. This analysis shows that for many outcomes, we can confidently reject effect sizes for similar policies estimated in the literature.

We also study the KAPET program using panel data on medium and large manufacturers, from Indonesia's Industrial Survey. Importantly, we find that firms in KAPET districts paid significantly lower sales, license, building, and land taxes than similar firms in non-treated districts. This is reassuring, given that the KAPET program was designed explicitly to reduce formal taxes, and our effect sizes are similar to policy intentions. In spite of these tax cuts, we find no impacts on firms' use of labor, output, value added, or productivity. Our quantitative null results validate other analyses of the KAPET policy, many of which are qualitative (e.g. Rahardja et al., 2012; Temenggung, 2013).

There are many possible reasons for the KAPET program's lack of success. The program began around the time of the Asian Financial Crisis and its subsequent political upheaval, so firms may have been dissuaded from using the incentives. Moreover, because the program operated at the beginning of Indonesia's decentralization program, heterogeneity in district administration may have undermined its effects (Soenandar, 2005). However, our findings are also consistent with literature suggesting that policies encouraging firms to locate in lagging regions may suffer from fundamental challenges, because those locations often have poor market access, unfavorable natural amenities, or low quality infrastructure.

To evaluate the welfare effects of the KAPET program, and to understand whether it could have been redesigned to encourage a more productive reallocation of economic activity, we specify a quantitative spatial model of Indonesian economic geography, based on Zárate (2022) and Henkel et al. (2021). In the model, workers choose a location of residence and whether to work in the formal or the informal sector. Between districts, trade and migration are both costly, and workers value local public goods, which are paid for by local tax revenues and transfers from the central government.

Firms use labor and capital to produce goods, and they face input wedges (Hsieh and Klenow, 2009), a portion of which includes taxes paid by formal sector firms. Reducing taxes on capital inputs can encourage sector switching and increase firm entry. If tax cuts target places with greater potential for agglomeration economies, they could reduce spatial misallocation and lead to overall welfare gains. At the same time, tax cuts in poorer districts with unattractive local fundamentals may not increase entry, and they could also constrain financing for local public goods, which reduces workers' welfare. After calibrating the model, we simulate the impact of the KAPET program, and we conduct a series of policy counterfactuals to study how it might have been redesigned.

We first demonstrate that our reduced form estimates are similar to model-based analogs, using a counterfactual scenario where we remove the KAPET tax cuts entirely. Then, using the model to evaluate the KAPET program, we find very small effects: the program only increased national welfare by 0.01 percent and output by 0.09 percent. Although it increased KAPET districts' formal sector employment by 0.62 percent, it also displaced informal workers and largely reshuffled economic activity without leading to growth. Because the KAPET program targeted districts that were poor and net recipients of central transfers, the tax cuts did not have a large impact on equilibrium local public goods.

Second, we find that the growth and welfare benefits of the KAPET tax cuts would have increased dramatically if they were combined with other local improvements that increased productivity and amenities, such as infrastructure that promotes communication or reduces commuting costs. This echoes findings from the literature on place-based policies suggesting that without other policies, tax incentives alone may be insufficient to stimulate investment (Duranton and Venables, 2021).[1]

Next, we use the model to evaluate alternative regional policies. After simulating separate 50 percent capital tax cuts in every district in Indonesia, we find that fiscal incentives in larger, denser districts would have been far more welfare and growth enhancing than tax cuts in poorer areas with larger informal sectors. This finding supports ideas in the literature that firms tend to locate in cities that are too small from the perspective of maximizing social welfare (Henderson, 1974; Albouy et al., 2016).

Finally, we evaluate other place-based policies that could have targeted KAPET districts. Similar to our baseline findings, cutting labor taxes or reducing overall misallocation wedges alone would not substantially increase output or welfare. Instead, reducing formal entry barriers by promoting greater competition would lead to significant growth and welfare improvements. We also find that despite a long history of using fiscal transfers to improve local public goods (Lewis, 2001), increasing transfers to KAPET districts would have increased migration to those areas, but it would also encourage people to stay in less attractive places, reducing welfare (Henkel et al., 2021).

Literature Contribution. This paper contributes to a large empirical literature evaluating the benefits and costs of different place-based policies (Neumark and Simpson, 2015). Much of this literature focuses on programs implemented in the U.S. and Europe and finds decidedly mixed results.[2] We also contribute to a growing literature that studies these policies in developing countries, and our work helps contextualize the findings of some reduced form papers in the literature.

[1] On the need to combine fiscal incentives with other policies in designing place-based policies or special economic zones, see also Farole (2011), Frick et al. (2019), and Frick and Rodríguez-Pose (2023).

[2] In particular, the evidence on enterprise zones in the U.S., which are similar in some ways to KAPET zones, is quite mixed, with results that often differ depending on the authors' research design (c.f. Elvery, 2009; Ham et al., 2011; Lynch and Zax, 2011; Freedman, 2013).

For example, many place-based policies enacted in India and China are actually combinations of fiscal incentives with other complementary policies. Shenoy (2018) finds when combined with infrastructure improvements, tax incentives and investment subsidies in Uttarakhand, India led to large growth in nighttime lights. Gallé et al. (2024) find that a combination of fiscal incentives, administrative benefits, and a unifying legal framework in India's Special Economic Zone Act of 2005 led to significant employment and manufacturing growth. Lu et al. (2019) also find large employment, capital, and output effects of China's SEZs, which combine fiscal incentives with discounted land-use fees and special treatment in securing bank loans. Our results suggest that tax incentives alone are unlikely to be impactful, and that the success of policies studied in the literature owes not just to the fiscal incentives themselves, but also to the infrastructure improvements, administrative streamlining, or other policies with which they were combined.[3]

Additionally, we build on a growing literature that uses structural models to study the welfare consequences of place-based policies (Fajgelbaum and Gaubert, 2020; Gaubert et al., 2021). Several papers on developing countries examine the effects of migration costs (Bryan and Morten, 2019), international trade and migration restrictions (Tombe and Zhu, 2019), pollution induced migration (Khanna et al., 2021), and the spatial allocation of educational investments (Hsiao, 2024), but few have looked at the effects of capital taxation in such settings.

Our model reflects key features of developing countries and distinctive characteristics of Indonesia. Prior work has emphasized the importance of the informal sector as a source of employment in many LDC settings (e.g., Hsieh and Olken, 2014; Rothenberg et al., 2016), and we follow Zárate (2022) to allow for informal employment. Place-based policies that cut formal sector taxes may encourage more entry, leading workers to transition out of informality and into the formal sector. This sector switching could broaden the tax base and increase revenues, enhancing the ability of the national government to provide public goods. Our model also specifies taxes as one component of input market distortions embedded in production wedges, as in Hsieh and Klenow (2009).

At the same time, tax cuts may instead result in lower tax revenue if the sector switching effects are not pronounced, and the revenue base on which to levy a reduced tax rate does not grow. Because tax revenues are pooled by Indonesia's unitary central government and distributed to districts through a system of regional transfers, we follow Henkel et al. (2021) to formulate this transfer system, but do so in a model with both trade and migration frictions.

Finally, we incorporate capital as an input for firms, which was the explicit focus of the KAPET policy. In most quantitative economic geography models, firms only use labor for production (Allen and Arkolakis, 2019; Bryan and Morten, 2019), but we allow for capital supplied elastically on a national market, as in Khanna et al. (2021). Our general equilibrium evaluation of the KAPET program also complements recent work that adopts a reduced form, partial equilibrium approach to study the equity and efficiency of place-based policies (Siegloch et al., 2025). The benefit of our general equilibrium modeling is to allow for the study of rich policy counterfactuals and non-marginal policy changes.

2. Regional policies in Indonesia and the KAPET program

Spatial differences in the concentration of economic activity, employment, and output have long been a feature of Indonesia's economy. Some differences owe to natural resources, but in the early 17th century, the Dutch East India Company practiced an uneven development strategy, promoting extractive enclaves in the form of plantations and mines in Sumatra, Sulawesi, and Kalimantan while encouraging more balanced, diversified growth in Java (Hill, 2000). Regional disparities in output and income have also strained internal relations throughout Indonesia's history.[4]

After independence, Suharto's New Order regime (1967–1998) enacted several policy initiatives to combat regional inequality (Soenandar, 2005). For example, Suharto's first five-year development plan introduced several investment incentives to promote growth in the Outer Islands, including a tax holiday for firms locating outside of Java.[5] Despite these efforts, regional inequalities have remained considerable. For example, in 1990, non-oil GDP per capita in the richest district (Kediri, in East Java) was more than 70 times that of the poorest district (Lampung Barat, in South Sumatra).

KAPET. After the mid 1980s when off-Java tax breaks were eliminated, efforts to promote growth in Indonesia's poorer regions were limited until the creation of the Integrated Economic Development Zones (*Kawasan Pengembangan Ekonomi Terpadu*, or KAPET). The KAPET program, first announced in 1996, was designed to accelerate growth and development in Eastern Indonesia, which consists of the island groups of Kalimantan, Sulawesi, Maluku, Papua, and Nusa Tenggara. By law, formal businesses locating in KAPET zones were eligible for several incentives that reduced corporate income tax. These measures included a partial tax holiday, with a 30 percent reduction of taxes on capital, expanded choices for depreciation and amortization of capital and losses, fiscal loss compensation for 10 consecutive years, and lower income taxes on dividends for foreign taxpayers.[6] Other benefits included: (1) a corporate income tax exemption on imports of capital goods, raw materials, and other equipment; (2) a 50 percent reduction on building and land taxes for both new construction and expansion; and (3) additional benefits, including being able to deduct 50 percent of employee costs from corporate taxable income (Presidential Decree 9/1998, Article 22).

By using fiscal incentives, policymakers hoped that KAPET areas would attract investment and stimulate manufacturing and export growth. Planners also intended to promote growth and development for micro, small, and medium-sized enterprises (MSMEs) to reduce the size of the informal sector. Growth in formal manufacturing would stimulate labor demand, creating jobs, raising wages, and reducing poverty and unemployment.

Presidential Decree No. 89 of 1996 created the KAPET program, building on earlier programs aimed at accelerating development in Eastern Indonesia. Apart from one KAPET zone created in Papua (which we ignore in our analysis because of data quality concerns), 12 KAPET zones were launched by presidential decree in January 1998, before Suharto resigned, and in September 1998, after Habibie became president. Table 1 lists the names and locations of the 12 KAPETs in our sample, the presidential decree that established them, and the year they started operation. Although one KAPET was established in Banda Aceh, in North Sumatra, the remaining 12 other KAPETs were located in Eastern Indonesia.

On paper, the KAPET program can primarily be thought of as a cut to taxes that firms pay on capital inputs. However, in practice, the program suffered from several implementation issues. First, many

[3] See also Wang (2013) on China's SEZs, or Koster et al. (2019) on the opening of science parks in Shenzhen.

[4] For example, in the 1950s, violent separatist movements in Kalimantan and Sumatra, notably in Aceh, threatened to overwhelm Sukarno's early presidency (1945–1967) and dissolve the Indonesian nation.

[5] For an in-depth discussion of the history of regional policies and investment incentives in Indonesia, as well as some background on the KAPET program, see Temenggung (2013). Some policies that expanded in the late 1970s and early 1980s, such as Indonesia's Transmigration program, were also designed in part to stimulate economic activity in unpopulated, underdeveloped regions (Bazzi et al., 2016).

[6] Indonesian corporate taxpayers pay both corporate income tax, which apply to profits, and value added taxes, which are levied at a flat rate of 10 percent, with rebates for exports (Basri et al., 2021). The KAPET incentives mostly applied to corporate income taxes and not to value added taxes.

Table 1
KAPET locations.

Name	Province	# of Districts	Year	Pres. Decree #	Well Implemented
KAPET Batulicin	Kalimantan Selatan	1	1998	Keppres 11/1998	Yes
KAPET Sasamba	Kalimantan Timur	3	1998	Keppres 12/1998	
KAPET Sanggau/Khatulistiwa	Kalimantan Barat	3	1998	Keppres 13/1998	Yes
KAPET Minado-Bintung	Sulawesi Utara	3	1998	Keppres 14/1998	Yes
KAPET Mbay	Nusa Tenggara Timur	1	1998	Keppres 15/1998	
KAPET Pare-Pare	Sulawesi Selatan	5	1998	Keppres 164/1998	Yes
KAPET Seram	Maluku Tengah	1	1998	Keppres 165/1998	
KAPET Bima	Nusa Tenggara Barat	2	1998	Keppres 166/1998	
KAPET Batui	Sulawesi Tengah	1	1998	Keppres 167/1998	
KAPET Bukari/Bank Sejahtera Sultra	Sulawesi Tenggara	3	1998	Keppres 168/1998	Yes
KAPET DAS Kakab	Kalimantan Tengah	3	1998	Keppres 170/1998	
KAPET Sabang/Bandar Aceh Darussalam	Propinsi D.I. Aceh	2	1998	Keppres 171/1998	Yes

Source: BAPPENAS and various presidential decrees (*Keputusan Presiden*, or *Keppres*). This list differs slightly from Soenandar (2005), who does not include KAPET Sabang in his analysis. In our empirical work, because of concerns about data quality, we drop Papua and KAPET Biak entirely from the analysis. The number of districts is based on 1990 district definitions. The determination of "well implemented" KAPET zones follows Soenandar (2005) and Financial and Development Supervisory Board (BPKP) (2003).

KAPET zones were created in early 1998, right before the Asian Financial Crisis and its subsequent political upheaval. Despite the tax breaks offered by the program, these contemporaneous political and macroeconomic events may have dissuaded investors from the region, especially during the early years of the program. However, in 2000, Regulation No. 20 offered some streamlining of the KAPET incentives and more flexibility to firms who would use them, and this signals the government's intent to continue operating the program after the macroeconomic and political upheaval.

A second issue is that KAPETs were managed under a decentralized structure, and local governments had substantial authority over how to use program funds. In the early days of Indonesia's "big bang" decentralization, there was considerable uncertainty over administrative authority and obligations. Local governments had trouble coordinating with the central government and often lacked the capacity to manage and implement many programs transferred to them (Hofman and Kaiser, 2004).

According to a report conducted by the executive director of the Development Council for the Acceleration of Eastern Indonesia and the Developing Body of KAPET, by 2003, only 6 KAPET zones had actually established transparent fiscal incentives for firms: (1) Sabang/Bandar Aceh; (2) Sanggau/Khatulistiwa; (3) Batulicin; (4) Minado-Bintung; (5) Pare-Pare; and (6) Bukari/Bank Sejahtera Sultra (Financial and Development Supervisory Board (BPKP), 2003, cited in Soenandar, 2005). The remaining KAPETs struggled to do so because of coordination issues between the center and local administrative authorities. Table 1 lists this set of "well implemented" KAPETs, which we take as our main group of treated locations. However, for robustness, we also consider the full set of KAPET zones.

Fig. 1 presents a map of the districts treated by the program. Although 28 different districts were nominally treated by the program, it was only properly implemented in 17 districts.[7] Districts were selected for the KAPET program according to several different criteria: (1) favorable geography; (2) potential for economic growth; (3) the presence of leading sectors capable of boosting growth in hinterland areas; and (4) potentially large investment returns. Each KAPET zone was established in a different province. Within provinces, it was hoped that KAPET zones would be growth centers, attracting activity from hinterland areas.

[7] Note that based on legislation documents, 21 districts were fully treated by the program, but 7 districts were only partially treated, with incentives that just covered specific sub-districts. Because much of our data is coded at the district level, we ignore differences between fully treated and partially treated districts in our empirical work.

3. Data

In this section, we first define districts, the spatial unit of analysis used in our empirical work. Then, we describe several key datasets used in the paper: (1) geographic characteristics and satellite data; (2) census data to measure demographic outcomes; and (3) data on economic and industrial activity.[8]

Districts. Throughout the paper, we use Indonesia's districts (*kabupaten*) as the primary spatial unit of analysis. The district is the second administrative division in Indonesia, nested below the province, but above the subdistrict (*kecamatan*) and village (*desa* or *kelurahan*). Because many districts were divided and partitioned into new districts during the process of decentralization after the fall of Suharto, we aggregate the data back to 1990 definitions in order to achieve a consistent geographic unit of analysis.[9] Our sample contains 290 districts, with a median land area of 1,877 square kilometers. This is similar to the size of U.S. counties, which have a median area of 1,595 square kilometers.[10]

Geographic Characteristics. We use the Harmonized World Soil Database (HWSD) and other sources to measure many agroclimatic features. These include measures of topography, such as elevation, slope, ruggedness, soil characteristics, and distance to rivers and the coast. We also construct measures of centroid distance to Jakarta, distance to major roads, and distance to major ports to proxy for market access. These variables capture aspects of districts' natural amenities that may influence productivity, and they also proxy for the "favorable geography" criterion used to govern selection into treatment.

Demographic Characteristics. For the last 40 years, Indonesia's Central Statistical Agency (*Badan Pusat Statistik*, or BPS) has collected high quality decennial population censuses. We use IPUMS extracts of the 1990 Census and unit-level data from the 2000 and 2010 Censuses to measure district-level demographic characteristics. Our data includes the size of the total population, the percent of the population with different levels of educational attainment, and measures of migration flows.

We also use data from the World Bank's Indonesia Database for Policy and Economic Research (INDO-DAPOER), which contains population and social indicators at the district-level, including annual data on district populations as well as measures of infrastructure.

[8] More information on these variables, including precise details on how various outcome measures were calculated, can be found in Appendix B.

[9] Until the late 1990s, district boundaries were relatively stable (Booth, 2011). However, because of the decentralization process, many subdistricts split off from their original districts, forming new districts. The number of districts increased from 302 in 1999 to 514 in 2014, through a process known as *pemekaran* or blossoming (Bazzi and Gudgeon, 2021).

[10] Indonesia's major cities (*kota*) are also given separate district identifiers, and these designations are also used in the analysis.



Fig. 1. Map of treated districts.
Notes: Authors' calculations using BPS district shapefiles. Information on treated districts was taken from the text of various presidential decrees, listed in Table 1. Here, we assume that districts with poorly implemented KAPET programs were not treated. In our empirical work, because of concerns about data quality, we drop Papua and KAPET Biak entirely from the analysis; these areas are not shown on the map.

Output, Expenditure, Employment, and Industrial Activity. We use two different indicators to measure economic activity at the district level: (1) annual data on real non-oil gross domestic product from sub-national product accounts data produced by BPS; and (2) annual data on nighttime light intensity from the National Oceanic and Atmospheric Administration (Henderson et al., 2012). Although light intensity may not be a good proxy for economic activity in rural areas, it has been shown to predict economic activity in urban settings (Gibson et al., 2021).

We also use various waves of the national socioeconomic survey (*Survei Sosial Ekonomi Nasional* or *Susenas*) to measure the total consumption expenditures of households and the fraction of the population that is classified as poor, according to BPS definitions.[11] We combine these measures of economic activity with employment estimates from the National Labor Force Survey (*Survei Angkatan Kerja Nasional* or *Sakernas*). The *Sakernas* provides annual, district-level estimates of the fraction of workers who are employed, and who work in the formal and informal sectors. Following Comola and De Mello (2011) and BPS, we measure the share of employed workers who work in the informal sector as the share of workers who are self-employed or family/unpaid workers.[12]

Finally, we use firm-level data from an annual plant-level survey of medium and large firms, the Survey of Manufacturing Establishments (*Survei Tahunan Perusahaan Industri Pengolahan*, or SI). The SI is intended to be a complete annual enumeration of manufacturing plants with 20 or more employees. The survey records information on plant employment sizes, their industry of operation, cost variables, and measures of output and value added. Capital is measured as the firm's reported book value of capital. We construct district-level aggregates of the SI data, but we also use the annual firm-level panel structure of the data in our analysis below.

3.1. Pre-treatment summary statistics

Table 2 presents summary statistics of several variables, each measured before the KAPET program was launched in 1998. The first set of columns reports the means, standard deviations, and sample sizes of variables for districts where the KAPET program was well implemented; we drop poorly implemented KAPET districts from this analysis.[13] The second set of columns reports the difference in means between treated and "all non-treated" districts, where this sample includes all districts in Indonesia that were not selected by the KAPET program. Finally, in the third set of columns, we report the difference in means between treated and non-treated districts in the Outer Islands. This final comparison excludes districts on Java and Bali which were not targeted by the policy.

Panel A reports summary statistics for geographic characteristics. Compared to all non-treated districts, KAPET districts are more rugged and farther from Jakarta. While some geographic differences are statistically significant, differences tend to shrink and significance levels fall when comparing treated districts to non-treated counterparts in the Outer Islands. In Panel B, we compare districts on the basis of economic activity. In 1993, average GDP was smaller in treated districts, as expected given that policymakers were targeting poorer areas for the program. Relative to all non-treated districts, treated districts also experienced significantly lower intensity of nighttime lights in 1992 as well as significantly smaller pre-treatment growth in nighttime lights from 1992–1997. While treated districts had lower employment rates, a greater share of informal employment, and fewer manufacturing firms, these differences are not always significant. Again, when we restrict our comparison to non-treated districts in the Outer Islands, these differences tend to attenuate and lose significance.

In Panel C, we show that relative to all non-treated districts, KAPET districts were less populated in 1990, but there were no significant educational differences between treated and non-treated districts. However, these differences grow smaller when comparing treated to non-treated districts within the Outer Islands. Panel D also shows that there were no significant differences in the quality of public goods and infrastructure between treated and non-treated districts. Overall, the results shown in Table 2 emphasize negative selection of KAPET districts. They also highlight that KAPET zones in the Outer Islands may provide a more suitable comparison group to KAPET districts, allowing us to better address first order concerns about endogenous program placement.

[11] Official poverty lines use thresholds for total expenditures per capita that differ by rural and urban areas. We use official poverty lines from BPS, reported by Priebe (2014).

[12] Only workers who work for a salary are considered to work in the formal sector.

[13] Results comparing all KAPET districts in this analysis, regardless of implementation, are similar and available on request.

Table 2
Summary statistics: pre-treatment characteristics.

			All		Outer Island	
	Treated		Non-Treated		Non-Treated	
Panel A: Geographic Characteristics	Mean (sd)	N	Δ Mean	N	Δ Mean	N
Average elevation (meters)	272.12 (208.25)	17	−32.38	252	−57.43	136
Vector ruggedness measure	0.28 (0.07)	17	0.05***	252	0.01	136
Distance to Jakarta (km)	1525.38 (450.16)	17	676.60***	252	331.61**	136
Distance to nearest river (km)	5.12 (4.68)	16	0.84	252	−0.57	136
Distance to nearest major port (km)	88.91 (86.36)	17	−5.75	248	−20.93	135
Distance to nearest major road (km)	0.08 (0.05)	17	0.01	252	−0.02	136
Kota Status (0 1)	0.24 (0.44)	17	0.03	252	0.04	136
Panel B: Economic Activity						
District GDP in 1993 (constant prices, Rp. mil.)	525.60 (461.28)	17	−533.78***	252	−32.14	136
Percent Change in GDRP (1993–1996)	0.24 (0.07)	17	0.01	252	−0.01	136
Average nighttime light intensity, 1992	1.17 (2.44)	17	−4.38***	252	−1.17	136
Change in average light intensity, 1992–1997	0.00 (0.00)	17	−0.00***	252	−0.00	136
Change in any night lights, 1992–1997	0.14 (0.15)	17	−0.10**	252	0.02	136
Employment rate (1990, Sakernas)	0.61 (0.12)	16	−0.06*	248	−0.06*	134
Informal employment share (1990, Sakernas)	0.79 (0.14)	15	0.06	248	−0.00	134
Number of medium firms (1990, SI)	7.29 (7.57)	17	−38.15***	252	−5.93**	136
Number of large firms (1990, SI)	3.29 (7.32)	17	−13.10***	252	−1.63	136
Number of workers (1990, SI)	2280.84 (5754.16)	17	−6782.40***	252	−457.05	136
Median wage per worker, SI 1990 (Rp '000)	984.47 (497.85)	16	209.07	230	38.57	114
Panel C: Demographic Characteristics						
Total population, 1990 (thousands)	412.93 (276.29)	17	−275.40***	252	−29.17	136
% of pop. w/ high school completion, 1990	4.13 (2.70)	16	0.22	252	0.25	136
% born in same province, 1990	0.87 (0.12)	16	−0.04	252	−0.02	136
% in same province 5 years ago, 1990	0.96 (0.06)	16	−0.01	252	−0.01	136
Panel D: Public Goods and Infrastructure						
% of HH w/ access to electricity, 1996–1997	66.00 (18.31)	17	−4.91	252	6.58	136
% of HH w/ access to safe water, 1996–1997	36.78 (18.68)	17	−2.18	252	3.17	136
% of HH w/ access to phone lines, 1996	3.70 (3.40)	17	−0.40	252	−0.01	136
% of HH w/ access to safe sanitation, 1996–1997	55.09 (13.29)	17	4.94	252	4.07	136

Notes: Authors' calculations using data described in Appendix B. The Δ mean columns report the difference in means between treated and non-treated districts. In the first set of columns, all non-treated districts are used as the comparison group, while in the second set of columns, only non-treated districts excluding Java and Bali are used. The significance stars in this table are computed by conducting a two-sided equality of means *t*-test between comparison groups. */**/*** denotes significant at 10%/5%/1% levels.

4. Evaluating Indonesia's KAPET program

To study the impact of Indonesia's KAPET program on outcomes, we estimate parameters of the following two-way fixed effects (2WFE) regression equation:

$$y_{dt} = \alpha_d + \alpha_t + \beta T_{dt} + \left(\mathbf{x}_d \times t\right)' \theta + \varepsilon_{dt} \quad (1)$$

where d indexes districts, t indexes years, y_{dt} is an outcome variable for district d in year t, T_{dt} is a time-varying indicator for whether or not district d was treated by the KAPET program in year t, $\left(\mathbf{x}_d \times t\right)$ is a vector of predetermined controls (described below) interacted with a time trend, and ε_{dt} is an error term. The key parameter of interest is β, which measures the average treatment effect on the treated (ATT) on y for KAPET districts relative to non-treated districts.[14]

A major concern in assigning a causal interpretation to β in Eq. (1) is that T_{dt} was not randomly assigned. As discussed in Section 2, although program documents emphasize that districts were selected for their economic growth potential, the results in Table 2 suggest that KAPET districts were, on average, not very different from other districts in the Outer Islands. However, if policymakers hoped that the program would stimulate development in poorer areas, we might expect a naive comparison between treated and non-treated districts to result in a downward bias in estimates of β, even within the Outer Islands.

To address this potential bias, we adopt two approaches. First, we use propensity-score re-weighting, where the weights come from a first-step estimation of the probability of being selected for the KAPET program as a function of baseline control variables, $\mathbf{x}_d$. Specifically, in line with stated selection criteria, we include six geographic controls (the district's average elevation, average ruggedness, two measures of the share of each district with different levels of slope, distance to Jakarta, distance to the nearest major port), two measures of population (log total population in 1990, and the 1990 rank of the district's population in its province), and one baseline measure of economic activity (the 1990 share of the population working in manufacturing). We also include the change in the provincial vote total for Golkar (Suharto's political party) in legislative elections between 1997 and 1999. Changes in election results after the fall of Suharto may reflect political sentiment towards the ruling party in 1998 when decisions about KAPET locations were made.

Appendix Table A.1 reports propensity score estimates. Despite using only a parsimonious set of variables in $\mathbf{x}_d$, our propensity score explains a sizeable amount of cross-sectional treatment variation, with pseudo-R^2's between 0.37 and 0.46. We find that ruggedness, proximity to ports, distance to Jakarta, larger populations in 1990, and being a district with higher population ranks in the province were all associated with greater treatment likelihood. We also find that districts from provinces with a greater increase in Golkar's vote share were more likely to be treated. Appendix Figure A.1 plots a histogram of the propensity score across treated and all non-treated districts (Panel A) and across treated and non-treated districts in the Outer Islands (Panel B). Overall, this figure demonstrates that overlap improves with the Outer Islands comparison.

Using weights derived from the propensity score models estimated in Appendix Table A.1, we implement a reweighting estimator of the

[14] Note that when constructing T_{dt}, we assume that treatment begins in the year 1999. This choice is both due to potential data quality issues in 1998, which coincides with the official start date of the program and also the Asian Financial Crisis, and also because the program began in several KAPET districts in the latter part of the year. However, our results are also robust to using 1998 as the start year; these results are available upon request.

two-way fixed effects model in Eq. (1). This approach assigns greater counterfactual weight to non-treated districts with similar underlying natural advantages, pre-treatment economic development, and political characteristics. We additionally control for the $\mathbf{x}_d$ characteristics interacted with a time trend, allowing for trend differences in districts with different baselines.

While our 2WFE with propensity-score reweighting approach has attractive features, it assumes that treatment is strictly exogenous given the history of regressors. Given that we can only explain 37–46 percent of variation in treatment, this assumption may be unconvincing. Consequently, we also adopt a second method to address endogeneity, namely the synthetic differences-in-differences (SDID) approach of Arkhangelsky et al. (2021). SDID combines attractive features of traditional difference-in-differences (DID) estimators with the method of synthetic controls (SC) (Abadie et al., 2010; Abadie, 2021). As in the SC approach, the estimator matches pre-exposure trends for treated districts by reweighting non-treated district observations, weakening reliance on strong common trends assumptions. However, the method also allows for district fixed effects as in traditional DID estimators, and it is robust to more flexible models of policy targeting.[15] We fit a separate SDID model for each outcome variable and report more details and diagnostics in Online Appendix D.

Pre-Treatment Trends in Outcomes. A key concern with both approaches is that the change in outcomes for non-treated comparison districts may not be a valid counterfactual for what would have happened to treated districts if they had not been treated. To assess the common trends assumption, Fig. 2 uses an event study specification to study the evolution of average differences between treated and control districts along 4 key outcomes: (1) log employment; (2) log gross domestic regional product (GDRP); (3) light intensity; and (4) the log total number of firms.

For each outcome, we report results for two different estimators: (1) the 2WFE with propensity-score reweighting estimator; and (2) the SDID estimator.[16] The 2WFE graphs report 95 percent confidence intervals based on robust standard errors clustered at the district level. The SDID confidence intervals are based on a clustered bootstrap, where we independently resample districts and use 1000 bootstrap replications, following Arkhangelsky et al. (2021). Across outcome variables and estimators, all eight panels show no pre-treatment differences in outcomes before 1998—the omitted base year. These figures also show no significant differences afterwards, previewing our findings below.[17]

Average Treatment Effects on Treated Districts. In Table 3, Panel A, we provide estimates of β from Eq. (1) for growth in demographic and employment outcomes. The first row of Table 3 provides estimates of the ATT for log total population. Column 1 uses a naive two-way fixed effects estimator, while column 2 uses cross-sectional propensity-score weights, and column 3 additionally includes the $\mathbf{x}_d$ controls interacted with time trends. Column 4 reports an unconditional SDID estimator, and column 5 reports the SDID estimator with the $(\mathbf{x}_d \times t)$ controls.

Quantitatively, the point estimate in column 4 suggests that the KAPET program increased population growth by 0.6 percentage points $(= (e^{0.006} - 1) \times 100)$ relative to non-treated districts. Across specifications, the effects on population are small and vary in sign, but the results are not statistically significant, suggesting that the program had no treatment effect. We also obtain similar null results for the effect of the KAPET program on total employment (row 2). Despite some suggestive evidence that the program was associated with a modest increase in formal employment and a modest reduction in informal employment, across all 5 columns, no estimates are statistically significant. Row 5 also shows that there was no impact on log average earnings.

Panel B shows results of the KAPET program on manufacturing outcomes, using district-year aggregates from the SI. These estimates are much noisier than the estimates from Panel A, reflecting the fact that some districts only have a small number of large manufacturing firms. We see suggestive evidence that the program was associated with an increase in the number of formal firms, and we also see positive effects on output, capital, and wages for those firms, but the signs of these effects are not always robust across specifications, and they are seldom significant at conventional levels.

Finally, Panel C provides estimates of β for measures of regional output, economic growth, average consumption expenditures per household, and poverty rates. In general, we do not find any significant effects of the KAPET program on any growth outcomes, either as measured by regional product accounts data or using the nightlights data as proxies. Although we find that the program is sometimes associated with positive increases in expenditure per capita and reductions in poverty, coefficient signs are not always consistent, and effects are not statistically distinguishable from zero.

Simulated Power Analysis and Effect Size Comparison. The results in Table 3 fail to detect any significant effects of the KAPET program on a broad range of outcomes, including: (1) measures of demographic change; (2) employment; (3) manufacturing activity; (4) regional growth; and (5) consumption expenditures and poverty. However, in presenting these null results, several important questions remain. First, do they reflect a lack of statistical power? To quantify the ability of our research design to detect significant effects, we conducted a simulated power analysis, following Black et al. (2022). Second, are the confidence intervals implied by Table 3 small enough to reject that the KAPET program had similar effect sizes to other studies in the place-based policy literature? To address this question, we compare our estimates to influential studies of several prominent place-based policies in developing countries, including: (1) special economic zones in India (Chaurey, 2017; Shenoy, 2018; Hasan et al., 2021; Gallé et al., 2024); (2) special economic zones in China (Lu et al., 2019); and (3) regional transfers and investment subsidies to less developed regions of West Germany (Ehrlich and Seidel, 2018).

Fig. 3 compares several of the SDID estimates from Table 3, column 4, to similar treatment effects reported in the place-based policy literature. The first row shows the point estimate and 95% confidence interval for the impact of the program on log population. We also plot estimates of the long run population effects of a large-scale transfer program in West Germany designed to stimulate post-war development in an area near the Soviet border, known as the Zonenrandgebiet (ZBG). This estimate (grey triangle) is from Ehrlich and Seidel (2018) who use a spatial regression discontinuity design. Finally, in the first row of Fig. 3, we plot an estimate of the average population growth effects of special economic zones in India from Gallé et al. (2024), who also use a spatial difference-in-difference approach (grey square). Overall, the estimated confidence interval clearly rules out both effect sizes.

[15] For example, the SDID model can accommodate the following interactive fixed effects model, which is more general than the two-way fixed effects model presented in Eq. (1):

$$y_{dt} = \gamma_d' \lambda_t + \beta T_{dt} + \left(\mathbf{x}_d \times t\right)' \theta + \epsilon_{dt}$$

where γ_d is a vector of latent district-specific factors, λ_t is a vector of latent time factors, and other variables are as defined above. We can write this model in matrix form, $\mathbf{Y} = \mathbf{L} + \tau\mathbf{T} + \theta\mathbf{X} + \mathbf{e}$, where $\mathbf{L} = \mathbf{\Gamma}\mathbf{\Lambda}'$ collects the latent factors. The identification assumption is that $\mathbb{E}\left[\mathbf{e} \middle| \mathbf{L}, \mathbf{T}, \mathbf{X}\right] = 0$, so that conditional on treatment, the time-varying controls, and the latent factors, the errors are mean zero. However, the treatment assignment process can be arbitrarily correlated with $\mathbf{L}$. This formulation allows both treatment assignment and outcomes to depend on a flexible national time trend interacted with unrestricted district-specific intercepts.

[16] For the 2WFE results, separate district and year effects are included in the regression, and we additionally control for baseline characteristics interacted with time trends. For the SDID specification, we do not include controls.

[17] Similar event-study results can be found for a larger set of demographic and employment outcomes in Appendix Figure A.2 and for manufacturing outcomes in Appendix Figure A.3.

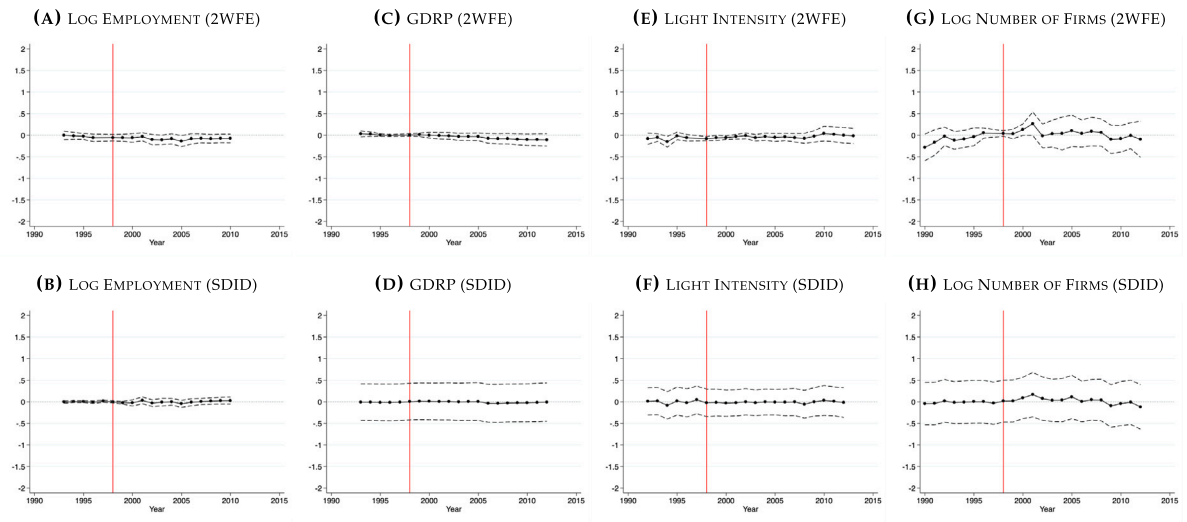


Fig. 2. Average differences Between KAPET and Non-KAPET districts.
Notes: This figure plots average differences between KAPET and non-KAPET districts in the Outer Islands using an event-study specification akin to the one described in Eq. (1). Each panel reports results for a different dependent variable. For the two-way fixed effects (2WFE) panels, non-treated districts are reweighted by $\hat{\kappa} = \hat{P}/(1-\hat{P})$, where $\hat{P}$ is the estimated probability that the district is a KAPET district. The 95 percent confidence intervals are reported using robust standard errors, clustered at the district level. For the synthetic difference-in-differences (SDID) panels, the 95 percent confidence intervals are based on a clustered bootstrap, where we independently resample districts and use 1000 bootstrap replications, following Arkhangelsky et al. (2021). The interaction between treatment and the year 1997 is left out of the specification, and the vertical line indicates the year 1998, when treatment began.

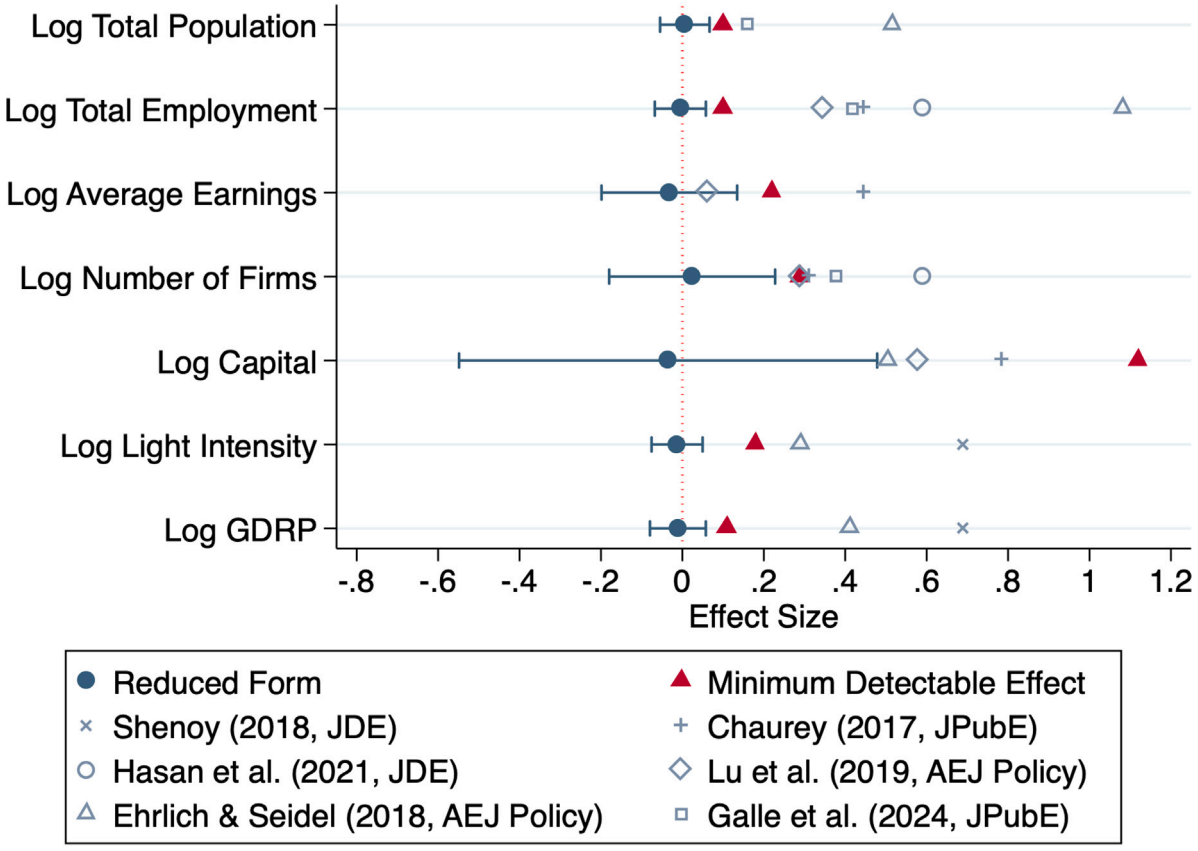


Fig. 3. Reduced form estimates: Simulated power and comparison with literature.
Notes: This figure compares the point estimates and 95% confidence intervals from our preferred SDID specification (Table 3, column 4) with both our simulated minimum detectable effect (MDE) sizes and also effect size estimates from the literature. We follow Black et al. (2022) to conduct a simulated power analysis and estimate MDEs for different outcome variables. A detailed description of the sources of the effect size estimates, as well as notes on comparability between outcomes, can be found in Appendix Table A.2.

While this is reassuring, perhaps our study is not sufficiently powered to detect such effect sizes, possibly due to a small number of treated observations or large sampling variation. Statistical power is defined as the probability that an estimate is found to be statistically significant when the true treatment effect is non-zero. Although power can be calculated directly for some specifications (e.g. Burlig et al.,

Table 3
Effects of the KAPET program: inner Island comparison.

Panel A: Demographics and Employment	(1)	(2)	(3)	(4)	(5)	Mean
Log Total Population	0.003	−0.027	−0.026	0.006	−0.018	12.788
	(0.026)	(0.035)	(0.037)	(0.031)	(0.030)	
Log Total Employment	−0.005	−0.054	−0.063	−0.005	−0.034	12.374
	(0.028)	(0.036)	(0.042)	(0.032)	(0.031)	
Log Formal Employment	0.107	0.034	0.010	0.094	0.081	10.730
	(0.128)	(0.136)	(0.133)	(0.081)	(0.075)	
Log Informal Employment	−0.027	−0.068	−0.070	−0.021	−0.042	12.073
	(0.042)	(0.046)	(0.052)	(0.035)	(0.037)	
Log Avg. Earnings (Sakernas)	−0.013	0.012	0.028	−0.032	0.021	12.053
	(0.065)	(0.077)	(0.067)	(0.085)	(0.084)	
Panel B: Manufacturing Outcomes	(1)	(2)	(3)	(4)	(5)	Mean
Log Number of Firms	0.073	0.104	0.117	0.024	0.013	2.254
	(0.112)	(0.154)	(0.143)	(0.104)	(0.100)	
Log Output	0.180	0.254	0.302	−0.010	0.201	3.181
	(0.242)	(0.252)	(0.292)	(0.246)	(0.238)	
Log Capital	0.108	0.469	0.481*	−0.035	0.250	2.457
	(0.245)	(0.296)	(0.271)	(0.262)	(0.251)	
Log Number of Workers	0.096	0.061	0.076	−0.032	−0.002	6.348
	(0.268)	(0.328)	(0.319)	(0.244)	(0.240)	
Log Wages Per Worker	0.106*	0.085	0.073	0.097	0.174*	0.829
	(0.059)	(0.069)	(0.095)	(0.094)	(0.095)	
Panel C: Regional Output and Expenditure	(1)	(2)	(3)	(4)	(5)	Mean
Log Light Intensity	0.000	0.032	0.023	−0.013	−0.004	0.684
	(0.047)	(0.051)	(0.032)	(0.032)	(0.034)	
Percent Any Light	0.013	0.014	0.009	−0.001	−0.004	0.188
	(0.020)	(0.025)	(0.014)	(0.011)	(0.012)	
Log GDRP	0.021	−0.065	−0.072	−0.011	−0.046	13.067
	(0.038)	(0.053)	(0.049)	(0.035)	(0.035)	
Log Avg. Expenditure Per Capita	0.004	−0.030	−0.029	0.005	0.015	10.907
	(0.033)	(0.041)	(0.040)	(0.031)	(0.033)	
Poverty Rate	−0.009	−0.001	0.001	−0.010	−0.020	0.350
	(0.015)	(0.018)	(0.017)	(0.017)	(0.018)	
N	3,726	3,726	3,726	3,726	3,726	
District FE	Yes	Yes	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	Yes	Yes	
Synthetic DiD	.	.	.	Yes	Yes	
Propensity-Score Reweighting	.	Yes	Yes	.	.	
$(\mathbf{x}_d \times t)$ Controls	.	.	Yes	.	Yes	
Placebo SE	.	.	.	Yes	Yes	

Notes: Each cell reports the coefficient from a regression of the given dependent variable (listed in the left-most column) on an indicator for whether or not the district was treated by the KAPET program in year t, after controlling for district and year fixed effects. Column 1 includes no other controls, while column 2 reweights non-treated district year observations by $\kappa = \widehat{P}/(1-\widehat{P})$, where $\widehat{P}$ is the estimated probability that the district is a KAPET district. Column 3 additionally adds the vector of $(\mathbf{x}_d \times t)$ controls. In columns 1-3, we report robust standard errors, clustered at the district level, in parentheses. Column 4 reports effects from an SDID estimator without controls, while column 5 uses an SDID estimator with $(\mathbf{x}_d \times t)$ controls. Standard errors in columns 4 and 5 are calculated with a clustered bootstrap, where we independently resample districts and use 1000 bootstrap replications, following Arkhangelsky et al. (2021). Sample sizes vary across outcomes but include as many as 3,726 observations. */**/*** denotes significant at 10%/5%/1% levels.

2020, for DID designs with clustered standard errors), it is difficult to do so in others. To assess power in our setting, we conduct a simulated power analysis, following Black et al. (2022). In this analysis, we first replace treated districts with resampled non-treated districts, and for those districts, we artificially introduce treatment effects of varying sizes. We then recalculate both the reweighted 2WFE and SDID estimators across 500 replications per effect size. Simulated power is calculated as the share of significant estimates. The minimum detectable effect (MDE) corresponds to the smallest effect size yielding 80 percent power at an $\alpha = 0.05$ significance level. We describe this procedure in more detail in Online Appendix D.

Fig. 3 reports the simulated MDEs for different outcomes as black triangles. For log population in the first row, the MDE is slightly larger than the upper bound of the confidence interval, but it is smaller than the two other log population effect sizes reported by Ehrlich and Seidel (2018) and Gallé et al. (2024). For this dependent variable, we conclude that our null result is fairly precise. Our 95 percent confidence intervals rule out effect sizes reported in prominent papers in the literature. The SDID estimator and our data structure have the power to reliably detect such effects, but strikingly, we do not.

The second row of Fig. 3 shows that our estimates of the effects of the KAPET program on log total employment are considerably smaller than other employment effects found in the literature, including Lu et al. (2019) on China's special economic zones, several studies of India's special economic zones (Chaurey, 2017; Shenoy, 2018; Hasan et al., 2021; Gallé et al., 2024), and a study of the West German ZBG (Ehrlich and Seidel, 2018). Our minimum detectable effect on log employment is also much smaller than these estimates.

In the third row of Fig. 3, we conduct the same analysis for log average earnings. While our research design and estimates can rule out many effects from the literature, others, such as the modest wage increases associated with China's special economic zones found by Lu et al. (2019), are too small for us to detect. Similarly, because we are working with an annual survey of large manufacturers instead of an economic census, our effects on firm creation (row 4) and capital (row 5) are much less precisely estimated than they are for other district-year outcomes. Although we can confidently rule out some large effect sizes, we cannot rule out others. However, for measures of regional growth, including light intensity (row 6) and gross domestic regional product (row 7), our estimates rule out those from the literature, and we have the power to detect smaller effects than those estimated in prior work.

Overall, despite imprecise effects on large manufacturing firm outcomes, the results of Fig. 3 suggest that the null results reported in Table 3 are not because of low statistical power. We can also

confidently reject that the KAPET program had effect sizes similar to those from other place-based policy evaluations in a variety of different contexts.[18]

ATT Estimates: Robustness. When estimating the impact of the KAPET program, our preferred estimates include poorly implemented KAPET districts in the control group. Appendix Table A.3 shows that our null results are robust to ignoring these implementation distinctions.[19] Another concern is that districts adjacent to treated districts might also be treated by the program due to spillovers, and this could attenuate effect sizes. Appendix Table A.4 shows that our results are robust to dropping districts adjacent to the treated KAPET districts from the comparison sample.

In Appendix Figure A.1, where we plot histograms of the propensity score, there was evidence suggesting that many non-treated districts had a low probability of being treated. This may signal a potential violation of the overlap assumption (Busso et al., 2014). To address this, in Appendix Table A.5, we trim the sample based on the propensity score, dropping both treated districts with large predicted values of treatment ($\hat{p} > 0.63$) and non-treated districts with very low treatment probabilities ($\hat{p} < 0.01$). Our null results from 2WFE specifications are robust to this sample restriction.

Finally, Appendix Table A.6 shows that our SDID results are robust to including different forms of controls. Column 1 reports our baseline SDID results without controls (from Table 3, column 4). Column 2 includes as controls a simple interaction between the $\mathbf{x}_d$ characteristics used to estimate the propensity score and an indicator for whether or not the year is greater than 1999, allowing the effects of baseline characteristics to differ before and after the year the program begins. Column 3 is our SDID with controls estimate (from Table 3, column 5), which includes the interaction of the $\mathbf{x}_d$ characteristics with a time trend. Column 4 includes interactions of the $\mathbf{x}_d$ characteristics with the treatment indicator, while Column 5 allows for separate trends based on $\mathbf{x}_d$ in treated and control areas. Overall, these different sets of controls have little effect on estimated effect sizes or significance levels, mitigating concerns that locational differences are a source of bias.

4.1. Supplementary evidence

Panel Data Estimates for Large Formal Firms. Another way to evaluate the effects of the KAPET program is to study how it affected individual firms, using panel data on manufacturers from the SI. In Table 4, we report these panel data estimates, focusing only on firms in the Outer Islands. The first column reports fixed-effects least squares estimates on firm-level outcomes, conditioning on separate fixed effects for each year, 5-digit industry, and district. In column 2, we restrict the sample to include only "incumbent" firms that started producing before the program began. Finally, in column 3, we replace the industry and district effects with firm fixed effects, and we limit the sample to incumbent firms. In all cases, we use the district-level propensity score to reweight the sample.

The first row shows that the KAPET program was associated with significant decreases in taxes paid on sales, business licenses, building permits, and land for firms affected by the program. Because the dependent variable is in logs, using the value of −0.614 from column 2, the coefficient estimate implies that firms in KAPET districts only paid 54 percent of the taxes that were paid in non-treated districts.[20] This suggests that the KAPET program was associated with a sizeable tax reduction, similar to the 50 percent cut for building, land, and employment taxes as stated in policy documents. In a sense, this result can be viewed as a "first stage," corroborating that the program actually reduced tax burdens for formal firms. However, in the remaining rows, we do not detect any significant impact of these tax cuts on output, value added, employment, capital, wages, or measures of output and value added per worker.

Repeated Cross-Section Results on Labor Market Outcomes and Expenditures. In Appendix Table A.7, we also report treatment effect estimates on individual labor market outcomes, using repeated cross-sections of the *Sakernas*. Column 1 shows estimates for individual-level outcomes, including indicators for employment and employment by sector (formal, informal, agriculture, or manufacturing), as well as log earnings. In the specification, we include district and year fixed effects, as well as demographic controls (age, squared age, gender, and an indicator for high school completion). Column 2 uses district-level propensity score reweighting, and Column 3 additionally includes the $(\mathbf{x}_d \times t)$ controls. The results in column 3 suggest a slight reduction in overall employment, a slight increase in formal employment, a slight decrease in informal employment, and a modest increase in wages, but none of these effects are statistically significant.

Finally, Appendix Table A.8 reports estimates using repeated cross-sectional data at the household-level from the *Susenas*. For these specifications, we include a vector of household-level controls (age and squared age of the household head, gender of the household head, educational attainment of the household head, and household size). Again, we do not find any statistically significant effects of the KAPET program on poverty or consumption expenditures.[21]

Discussion. Our reduced form results have generally found that as implemented, the KAPET program did not increase migration, stimulate economic growth, increase wages, or improve productivity. This begs an important question: why was this place-based policy unsuccessful? One hypothesis is that because KAPET areas tended to have poor local fundamentals, the tax subsidy required to stimulate growth and development in those areas was insufficient. In Appendix Table A.11, we provide some suggestive support for this idea. Using treatment interactions and our reweighted 2WFE regression specification, we find that formal employment growth was significantly lower in districts that were farther from Jakarta. This suggests that a policy that targeted locations with better market access or more favorable geography might have been more successful.

On the other hand, perhaps the Indonesian government could stimulate more growth by targeting larger cities with tax breaks. Or, maybe the tax cuts would have been more successful if they were combined with infrastructure improvements that improved amenities, increased productivity, and reduced trade costs. Unfortunately, the reduced form variation does not allow us to test these different hypotheses. To better understand why the KAPET program was unsuccessful, and to determine whether a differently designed tax incentive program could have induced a more optimal configuration of economic geography, we specify a quantitative spatial general equilibrium model. The next section describes this model, and the following section explains how we take the model to the data.

[18] Appendix Figure A.4 repeats this exercise using the 2WFE estimator instead of the SDID estimator, using results and the specification reported in Table 3, Column 3. This figure generally shows wider confidence intervals and larger MDEs, but we typically retain the power to rule out the effects reported in the literature that we could using the SDID estimator.

[19] We can also drop the poorly implemented districts entirely, and when we do, we obtain similar results; these are available upon request.

[20] To obtain this, we simply exponentiate the estimate of $\beta = -0.614$ to obtain $\exp(\beta) = 0.541$.

[21] Our district year and individual or household-year results might be masking some heterogeneity in effects by skill. Appendix Table A.9 shows results of the KAPET program on individual-level employment outcomes for individuals by skill (high school completion), and Appendix Table A.10 uses household-level *Susenas* regressions but splits the sample by whether the head of household was high or low skill. Overall, we detect no significant differences of the KAPET program separately by skill, and we cannot reject that the effect of the KAPET program was similar between skill groups.

Table 4
Firm-level regressions.

	All Firms	Incumbent Firms	
	(1)	(2)	(3)
Log Taxes on Sales, Licenses, Buildings and Land	−0.625**	−0.614**	−0.548**
	(0.240)	(0.282)	(0.239)
Log Output	0.019	0.034	0.086
	(0.127)	(0.112)	(0.114)
Log Value Added	0.032	0.023	0.067
	(0.119)	(0.107)	(0.106)
Log Number of Workers	−0.057	−0.053	−0.013
	(0.073)	(0.056)	(0.045)
Log Capital	−0.044	−0.063	−0.050
	(0.051)	(0.040)	(0.031)
Log Wages per Worker	−0.035	−0.011	0.094
	(0.074)	(0.063)	(0.059)
Log Output per Worker	0.018	0.009	0.015
	(0.020)	(0.017)	(0.015)
Log Value Added per Worker	0.010	0.003	0.005
	(0.009)	(0.008)	(0.008)
N	71,586	59,920	59,920
Year FE	Yes	Yes	Yes
District FE	Yes	Yes	.
5-Digit Industry FE	Yes	Yes	.
Firm FE	.	.	Yes

Notes: Each row reports the coefficient estimate of being treated by the KAPET program in year t on the outcome listed in the left-most column, using firm-level panel data for analysis. The first column uses all Outer Island firms to estimate effects, while columns 2 and 3 only focus on incumbent firms, which were in existence before 1998, when the program took place. All columns reweight non-treated firm-year observations by $\kappa = \hat{P}/(1-\hat{P})$, where $\hat{P}$ is the estimated probability that the firm's district is a KAPET district. Sample sizes vary across outcomes but include as many as 71,586 firm-year observations districts in column 1, and 59,920 firm-year observations in columns 2 and 3. Robust standard errors, clustered at the district level, are reported in parentheses. */**/*** denotes significant at the 10%/5%/1% levels.

5. A quantitative spatial model

Our model is based on Zárate (2022), which incorporates formal and informal sectors into a quantitative spatial model, and on Henkel et al. (2021), which allows for regional transfers to finance the provision of local public goods. The economy consists of N discrete locations (districts). Each district has two sectors, indexed by $z \in \{f, \text{inf}\}$, where f denotes the formal sector and "inf" denotes the informal sector.

Each sector in each district has different levels of amenities and productivity, both containing exogenous and endogenous components. Trade and migration are both costly, as in Allen and Arkolakis (2019). Workers are born in a location and decide where to live and a sector in which to work to maximize their utility. Firms use both labor and capital to produce output, and each input has a separate production wedge. Tax cuts on capital inputs, like those offered by the KAPET program, reduce production costs and output prices in the formal sector. However, tax cuts also reduce funding for local public goods, which workers value.

5.1. Workers and migration flows

The economy has a fixed total population of workers, given by $\overline{L}$. Workers are born in district i (origin), and they choose a district j in which to live (destination) and a sector z in which to work. Each district is endowed with an initial, exogenous stock of labor given by L_i^0, and we assume that $\sum_i L_i^0 = \overline{L}$. The total population of workers who choose to work in sector z in residence district j is denoted by L_{jz}.[22]

When making location and sector choices, workers have preferences over private consumption goods, local public goods, and amenities. These desirable features of locations are discounted by migration costs they incur when moving from their initial (birth) location. We assume a Cobb–Douglas form for the indirect utility of worker v who is born in district i, moves to district j, and works in sector z, given by:

$$V_{ijz}(v) = \frac{u_{jz}}{D_{ij}} G_j^{\gamma} \left(\frac{w_{jz}}{\mathbf{P}_j} \right)^{1-\gamma} \epsilon_{ijz}(v) \quad , \tag{2}$$

where u_{jz} denotes the level of amenities in district j and sector z, D_{ij} measures the cost of migrating from i to j, w_{jz} denotes the nominal wage in district j and sector z, $\mathbf{P}_j$ is the consumer price index in j, G_j measures local public goods in j, and $\epsilon_{ijz}(v)$ is an idiosyncratic preference draw.[23] The parameter γ measures the elasticity of substitution between public and private goods.

Amenities for sector z in district j have both an exogenous and an endogenous component:

$$u_{jz} = \overline{u}_{jz} L_j^{\beta} \quad , \tag{3}$$

where $\overline{u}_{jz}$ measures pre-determined local fundamentals that influence amenities (e.g., climate, distance to the coast, soil characteristics), and L_j measures the size of the local population, $L_j \equiv L_{j,f} + L_{j,\text{inf}}$, which is determined in equilibrium. The elasticity, $\beta < 0$, measures the strength of congestion forces, operating through various channels,

[22] Our model is static and assumes that individuals make a single sector and location choice.

[23] In a similar model, Henkel et al. (2021) assume that local public goods are given by $\widetilde{G}_j = G_j / L_j^{\kappa}$, where $\kappa \in [0, 1]$ governs the degree of rivalry in public goods. With this formulation, $\kappa = 0$ captures the case of pure, non-rival local public goods, while $\kappa = 1$ captures the full rivalry case. Our model assumes that local public goods are non-rivalrous and sets $\kappa = 0$.

including greater housing prices or longer commute times due to traffic congestion.[24] We allow local fundamentals, $\overline{u}_{jz}$, to differ across sectors in the same location. For instance, certain natural amenities may be valued more by formal sector workers (e.g. coastal access).

Finally, the idiosyncratic preference terms in Eq. (2), $\epsilon_{ijz}(v)$, are assumed to be independent draws from a nested Fréchet distribution, with shape parameters η and ν that govern the (inverse) dispersion of draws across locations and sectors, respectively.[25] As η decreases, workers' preferences are more dispersed across locations. This implies that they have stronger idiosyncratic attachment to certain locations and that migration is less responsive to changes in equilibrium location characteristics (e.g., wages, prices, public goods, or amenities). As ν decreases, workers' preferences are more dispersed across sectors, which suggests stronger attachment to one sector over another.

Each worker chooses the location and sector that offers the highest indirect utility. By properties of the nested Fréchet distribution, the probability that a worker from i moves to location j and works in sector z is given by:

$$\pi_{ijz} = \underbrace{\left\{ \frac{\left[\frac{1}{D_{ij}} G_j{}^{\gamma} \mathbf{P}_j^{\gamma-1} \Phi_j\right]^{\eta}}{\sum_{j'} \left[\frac{1}{D_{ij'}} G_{j'}{}^{\gamma} \mathbf{P}_{j'}^{\gamma-1} \Phi_{j'}\right]^{\eta}} \right\}}_{\pi_{ij}} \underbrace{\left\{ \frac{(u_{jz} w_{jz}{}^{1-\gamma})^{\nu}}{\sum_{z'} (u_{jz'} w_{jz'}{}^{1-\gamma})^{\nu}} \right\}}_{\pi_{ijz|j}}, \tag{4}$$

where $(\Phi_j)^{\nu} \equiv \sum_{z'} (u_{jz'} w_{jz'}{}^{1-\gamma})^{\nu}$ summarizes how attractive location j is to its residents. This choice probability can be expressed as the product of two probabilities: (1) location choice, π_{ij}; and (2) sector choice conditional on location choice, $\pi_{ijz|j}$.

We can measure the *supply of labor* to location j and sector z by summing migration flows from all origin locations i, as follows:

$$L_{jz} = \sum_{i \in N} \pi_{ijz} L_i^0 . \tag{5}$$

From the nested Fréchet distributional assumption, we can also show that the ex-ante average welfare of workers born in location i is given by:

$$\mathbb{E}\left[\max_{j,z} V_{ijz}(v)\right] = \Gamma\left(1 - \frac{1}{\eta}\right) \left\{ \sum_{j'} \left[\frac{1}{D_{ij'}} G_{j'}{}^{\gamma} \mathbf{P}_{j'}^{\gamma-1} \Phi_{j'}\right]^{\eta} \right\}^{\frac{1}{\eta}}, \tag{6}$$

where $\Gamma(\cdot)$ is the Gamma function.[26]

Demand for Goods. Workers have nested constant elasticity of substitution (CES) preferences over varieties of consumption goods produced by each sector in every location. In the first nest, consumers allocate their budget to spending across sectors $z \in \{f, \text{inf}\}$, and in the second nest, they choose varieties within sectors. Let ξ denote the elasticity of substitution between formal and informal goods, and let σ_z denote the elasticity of substitution between varieties in sector z. Following Zárate (2022), we assume that trade flows in the informal sector are more sensitive to trade costs, so that $\sigma_f < \sigma_{\text{inf}}$. The total quantity of consumer expenditures for an individual in location i, q_i, is given by:

$$q_i = \left(\sum_{z \in Z} q_{iz}^{\frac{\xi-1}{\xi}} \right)^{\frac{\xi}{\xi-1}} \quad \text{where} \quad q_{iz} = \left(\sum_{j \in N} \int_{\omega} q_{jiz}(\omega)^{\frac{\sigma_z - 1}{\sigma_z}} \, d\omega \right)^{\frac{\sigma_z}{\sigma_z - 1}}, \tag{7}$$

and where $q_{jiz}(\omega)$ measures the quantity of variety ω produced in location j and sector z that is consumed in location i. Location i's consumer price index, $\mathbf{P}_i$, is given by:

$$\mathbf{P}_i = \left(\sum_{z \in Z} \mathbf{P}_{iz}^{1-\xi} \right)^{\frac{1}{1-\xi}} \quad \text{where} \quad \mathbf{P}_{iz} = \left(\sum_{j \in N} \int_{\omega} p_{jiz}(\omega)^{1-\sigma_z} \, d\omega \right)^{\frac{1}{1-\sigma_z}}, \tag{8}$$

and where $p_{jiz}(\omega)$ measures the price of sector z variety ω produced in location j but consumed in location i. We also assume an iceberg trade technology, so that destination prices are equal to factory-door prices inflated by trade costs, i.e. $p_{jiz}(\omega) = \tau_{ji} p_{jz}(\omega)$ where $\tau_{ji} \geq 1$ measures trade costs from j to i.

5.2. Production

Each firm produces a differentiated variety, indexed by ω, in location i and sector z. All firms in the same sector and location are equally productive and compete monopolistically as in Krugman (1980). We assume that formal sector firms use both labor and capital for production, while informal firms only use labor.

Let F_{iz} denote the fixed entry cost that a firm in location i and sector z must incur in order to produce. When we calibrate F_{iz} below, we will show that $F_{i,f} > F_{i,\text{inf}}$ for all i, meaning that fixed entry costs are higher in the formal sector. We also assume that a large number of potential firms exist in each sector and location, ensuring that firm profits are driven to 0.[27] The production function for a representative firm in location i is given by:

$$y_{iz} = \begin{cases} A_{i,f} L_{i,f}^{\rho} K_i^{1-\rho} & z = f \\ A_{i,\text{inf}} L_{i,\text{inf}} & z = \text{inf} \end{cases}, \tag{9}$$

where A_{iz} measures productivity in location i and sector z, K_i is capital, and L_{iz} measures labor employed in sector z. Local productivity can be written as follows:

$$A_{iz} = \overline{A}_{iz} L_i^{\alpha} . \tag{10}$$

That is, productivity consists of an intrinsic part, $\overline{A}_{iz}$, which measures exogenous local fundamentals, such as natural amenities, and an endogenous part that is determined by the size of the local workforce, L_i. The constant elasticity, $\alpha > 0$, captures agglomeration externalities such as labor market pooling, input sharing, or knowledge spillovers (Rosenthal and Strange, 2004).

As in Hsieh and Klenow (2009), we assume that firms face production wedges on inputs, denoted by ψ_{iz}^L and ψ_i^K. These wedges include local taxes on labor and capital, denoted by t_i^L and t_i^K respectively, but they also reflect other sources of misallocation, such as subsidies, the cost of complying with regulations, or differential access to credit. The non-tax misallocation wedges for labor and capital are denoted by λ_{iz}^L and λ_{iz}^K respectively. The existence of misallocation wedges implies that the marginal revenue product of labor and capital may not be equalized across firms.

We assume that total production wedges impacting firms in location i and sector z are given by:

$$\psi_{iz}^L = \begin{cases} t_i^L + \lambda_{i,f}^L & z = f \\ \lambda_{i,\text{inf}}^L & z = \text{inf} \end{cases} \quad \text{and} \quad \psi_i^K = t_i^K + \lambda_i^K , \tag{11}$$

where t_i^L and t_i^K measure location i's formal tax rate on labor and capital inputs, respectively. The terms λ_{iz}^L and $\lambda_{i,f}^K$ also represent additional labor and capital misallocation wedges for firms in location i and sector z, respectively.

[24] We do not model housing directly, but this formulation is isomorphic to a model where consumers also demand housing, and where there is a constant elasticity of housing supply.

[25] The cumulative distribution function of the distribution is given by: $F(\vec{\epsilon}) = \exp\left\{ -\sum_{j \in N} \left(\sum_{z \in Z} \epsilon_{ijz}^{-\nu} \right)^{\frac{\eta}{\nu}} \right\}$.

[26] The average income of workers in location j is also given by $\overline{w}_j = \sum_z \pi_{jz|j} w_{jz}$.

[27] This is a similar setup to Dix-Carneiro et al. (2021), which assumes that firms enter the formal market endogenously.

We assume that informal firms do not pay local taxes and do not use capital in production. Because informal firms also tend not to comply with formal labor regulations, their overall production wedges are lower ($\psi^L_{i,\text{inf}} < \psi^L_{i,f}$), and we confirm this when calibrating the model in Section 6.

Factor Markets. The price of labor, w_{iz}, differs by location i and sector z, and is determined in equilibrium by local labor demand and supply. However, we assume that capital is supplied perfectly elastically on a nationally integrated capital market, and that the price of capital, $\bar{R}$, is constant everywhere. Absentee owners of capital collect and spend rental payments locally.[28] Due to differences in capital input wedges across locations, the effective rental price of capital in location i is given by:

$$r_i = \bar{R}(1+\psi^K_i)\,. \tag{12}$$

Pricing and Entry. Profit maximization implies that equilibrium prices for firms in location i and sector z are given by the usual constant markup over marginal production costs.[29] Prices in the formal sector and informal sectors are given by:

$$\begin{aligned} p_{i,f} &= \iota \cdot \left(\frac{\sigma_f}{\sigma_f - 1}\right) \frac{\left[w_{i,f}\left(1+\psi^L_{i,f}\right)\right]^{\rho}\left[\bar{R}\left(1+\psi^K_{i,f}\right)\right]^{1-\rho}}{A_{i,f}} \quad \text{and} \\ p_{i,\text{inf}} &= \left(\frac{\sigma_{\text{inf}}}{\sigma_{\text{inf}} - 1}\right) \frac{w_{i,\text{inf}}\left(1+\psi^L_{i,\text{inf}}\right)}{A_{i,\text{inf}}}\,. \end{aligned} \tag{13}$$

where ι is a constant given by $\iota \equiv \rho^{-1}(1-\rho)^{-(1-\rho)}$. If policymakers reduce formal input taxes, this translates into lower production costs which are passed onto consumers through lower goods prices. Because firms make zero profits, equilibrium output for firms is given by:

$$q_{iz} = \left(\sigma_z - 1\right) A_{iz} F_{iz}\,. \tag{14}$$

Since labor is employed by both types of firms in each location, the total numbers of formal and informal firms in location i are given by:

$$M_{i,f} = \frac{L^{\rho}_{i,f} K_i^{1-\rho}}{\sigma_f F_{i,f}} \quad \text{and} \quad M_{i,\text{inf}} = \frac{L_{i,\text{inf}}}{\sigma_{\text{inf}} F_{i,\text{inf}}}\,. \tag{15}$$

Given product demand, Eq. (7), we can show that the share of location j's expenditure on goods produced in location i and sector z is given by:

$$\zeta_{ijz} = \underbrace{\left(\frac{\mathbf{P}_{jz}^{1-\xi}}{\sum_{z' \in Z} \mathbf{P}_{jz'}^{1-\xi}}\right)}_{\zeta_{jz}} \underbrace{\left(\frac{M_{iz} p_{ijz}^{1-\sigma_z}}{\sum_{i' \in N} M_{i'z} p_{i'jz}^{1-\sigma_z}}\right)}_{\zeta_{ijz|z}}\,, \tag{16}$$

where ζ_{jz} denotes location j's share of expenditure in goods from sector z, and $\zeta_{ijz|z}$ denotes location j's expenditure share on goods from location i conditional on consuming goods from sector z. The term $\mathbf{P}_{jz} = \left(\sum_{i' \in N} M_{i'z} p_{i'jz}^{1-\sigma_z}\right)^{\frac{1}{1-\sigma_z}}$ is the equilibrium Dixit-Stiglitz price index.

Total revenue in location i and sector z is given by:

$$Y_{iz} = \sum_j \zeta_{jz} \cdot \zeta_{ijz|z} \cdot E_j\,, \tag{17}$$

where E_j measures total expenditure in location j.

5.3. Taxes, public spending, and fiscal transfers

In Indonesia, district governments use local taxes to raise some of revenue, but many districts also rely on transfers from the central government to finance local public goods (Lewis and Oosterman, 2009; Lewis, 2010).[30] Motivated by these features, our model assumes that local public goods are financed in three ways: (1) locally retained input wedges, $\lambda^L_{i,f}$, $\lambda^L_{i,\text{inf}}$, and $\lambda^K_{i,f}$; (2) local tax revenue from the formal sector, denoted by $\text{REV}_i = t^L_i w_{i,f} L_{i,f} + t^K_i \bar{R} K_i$; and (3) transfers of formal tax revenue from the national government, $\text{TRANS}_i = \theta_i \left(w_{i,f} L_{i,f} + \bar{R} K_i\right)$, which are applied to formal-sector GDP, and where θ_i is the net transfer rate (relative to total formal income). Note that θ_i is defined to be positive for recipient districts and negative for donor districts.

Given these assumptions, the sum of formal tax revenue collected by the central government from each district is given by $\sum_i \text{REV}_i$. The central government's total expenditure on public goods is given by $\sum_i (\text{REV}_i + \text{TRANS}_i)$. Note that if $\sum_i \left(\theta_i \times \text{TRANS}_i\right) = 0$, the central government budget is balanced. However, because θ_i is calibrated based on official statistics which may differ from model-based GDP, there may be a national deficit or surplus.

To ensure a balanced budget in the model, we assume the government's national transfer deficit or surplus, given by $\sum_i \text{TRANS}_i$, is paid to all districts in proportion to their total GDP:

$$\text{DEF}_i = -\chi_i \sum_i \text{TRANS}_i\,, \tag{18}$$

where χ_i is the fraction of district i's GDP in total national GDP, given by:

$$\chi_i = \frac{w_{i,f} L_{i,f} + \bar{R} K_i + w_{i,\text{inf}} L_{i,\text{inf}}}{\sum_i \left(w_{i,f} L_{i,f} + \bar{R} K_i + w_{i,\text{inf}} L_{i,\text{inf}}\right)}\,. \tag{19}$$

This essentially assumes that larger districts, because they contribute more to the transfer system, also have to play a greater role in financing the system's deficit. Given this setup, we can write total public goods supplied in district i as the sum of four terms: (1) formal sector tax revenues; (2) net transfers from the central government; (3) locally retained misallocation wedges; and (4) district i's share of the national transfer system deficit (or surplus):

$$G_i = \frac{\text{REV}_i + \text{TRANS}_i + \text{DEF}_i + \lambda^L_{i,f} w_{i,f} L_{i,f} + \lambda^K_i \bar{R} K_i + \lambda^L_{i,\text{inf}} w_{i,\text{inf}} L_{i,\text{inf}}}{\mathbf{P}_i}\,. \tag{20}$$

Aggregate local expenditure is therefore given by:

$$E_i = w_{i,\text{inf}} L_{i,\text{inf}} + w_{i,f} L_{i,f} + \bar{R} K_i + \mathbf{P}_i G_i \tag{21}$$

In our framework, a cut to local formal taxes will reduce productions costs and goods prices for formal-sector firms. It may also encourage informal firms to switch to the formal sector, potentially broadening the tax base and increasing funds to provide local public goods. At the same time, if sector switching effects are not substantial, local tax cuts may instead reduce the ability for districts to provide local public goods, both locally and everywhere else, because of fiscal spillovers operating through the transfer system.

[28] In contrast, under perfect capital mobility, the effective rental rate of capital, r_i, would be uniform across all locations.

[29] We can express the total cost of producing q units of variety ω in formal sector of location i as

$$C_{i,f} = \iota \cdot \left(F_{i,f} + \frac{q}{A_{i,f}}\right)\left[w_{i,f}\left(1+\psi^L_{i,f}\right)\right]^{\rho}\left[\bar{R}\left(1+\psi^K_{i,f}\right)\right]^{1-\rho}\,.$$

where ι is a constant defined in Eq. (13) below. Similarly, the total cost in the informal sector is given by:

$$C_{i,\text{inf}} = \left(F_{i,\text{inf}} + \frac{q}{A_{i,\text{inf}}}\right) w_{i,\text{inf}}\left(1+\psi^L_{i,\text{inf}}\right)\,.$$

.

[30] For instance, by the end of 2007, sub-national governments only accounted for about 8 percent of total public revenue (Lewis and Oosterman, 2009; Lewis, 2010). Because local governments have limited fiscal autonomy, the model abstracts away from any optimizing behavior or strategic considerations on the part of governments in setting tax or transfer rates.

Table 5
Calibrated parameters.

Parameter	Baseline Value	Source
Agglomeration Parameters α	0.056	Bryan and Morten (2019); Chauvin et al. (2017)
Congestion Parameters β	−0.04	Combes and Gobillon (2015)
Armington Elasticity of Substitution σ_z	7 for the informal sector 5.4 for the formal sector	Zárate (2022)
Sector Elasticity of Substitution between Goods ξ	2	Edmond et al. (2015)
Location Dispersion Parameter η	1.6	Tombe and Zhu (2019); Khanna et al. (2021); Fan (2019)
Sector Dispersion Parameter ν	1.95	Zárate (2022)
Elasticity of Trade Costs w.r.t Distance	−1	Head and Mayer (2014)
Expenditure Share on Public Goods γ	0.2	Henkel et al. (2021)
Rental Price of Capital $\bar{R}$	0.1	Hsieh and Klenow (2009)
Labor Share in Production ρ	0.81	Author's calibration from data

Notes: This table shows baseline values of parameters that we use to calibrate the model, as well as their associated sources.

5.4. Equilibrium

Given exogenous parameters $\{D_{ij}, \tau_{ij}, \bar{A}_{iz}, \bar{L}_{iz}, F_{iz}, \xi, \sigma_z, \gamma, \nu, \eta, \alpha, \beta, \rho\}$, the model's equilibrium is characterized by endogenous variables, $\{w_{iz}, \phi_{iz}, \mathbf{P}_{iz}, p_{iz}, L_{iz}, K_i\}$, which satisfy the following conditions: (1) goods market equilibrium, which states that total labor payments in each location equal the sum of that location's total sales to all locations in the economy; (2) labor market clearing, which is captured by location and sector choices in Eq. (5); (3) the government budget constraint, Eq. (20).[31]

6. Taking the model to the data

In this section, we explain how we calibrate key model parameters. When possible, we estimate these parameters with our data, but in some cases we choose values from prior work. In counterfactual simulations below, we explore the robustness of our results to different parameter values.

Parameters Calibrated from the Literature. Table 5 summarizes parameters that we calibrate from the literature and data. For our baseline, we choose the agglomeration elasticity, $\alpha = 0.056$, following Chauvin et al. (2017) and Bryan and Morten (2019). For the congestion elasticity, we work with a value of $\beta = -0.04$ from Combes and Gobillon (2015). We set the expenditure share of public goods to 0.2, following Henkel et al. (2021). For the Armington elasticity of substitution across varieties, we follow Zárate (2022) and set $\sigma_{\text{inf}} = 7$ and $\sigma_f = 5.4$.[32] We follow Edmond et al. (2015) and set the elasticity of substitution between formal and informal goods ξ to 2. For the Fréchet distribution governing idiosyncratic preferences, we set the location dispersion parameter η to 1.6, following Tombe and Zhu (2019), Fan (2019), and Khanna et al. (2021). We also set the sector dispersion parameter ν to 1.95, following Zárate (2022).

We follow Hsieh and Klenow (2009) and set the rental price of capital (excluding distortions) to 0.1, which combines a 5 percent real interest rate with a 5 percent depreciation rate. We calibrate the labor share in formal production using aggregate wage bill and revenue data from the 2006 Economic Census, finding that $\rho = 0.81$. Finally, for iceberg trade costs, we follow an extensive trade literature and set $\ln \tau_{ij}^{1-\sigma_z} = -1 \log \text{dist}_{ij}$ for both sectors z, where dist_{ij} measures the centroid distance between locations i and j.

Total Production Wedges. Following Hsieh and Klenow (2009), we back out total production wedges in each sector ($\psi_{i,f}^L$, $\psi_{i,f}^K$ and $\psi_{i,\text{inf}}$) using the ratio of wage bill and capital bills to total revenue as implied by profit maximization:

$$\frac{w_{i,f} L_{i,f}}{p_{i,f} q_{i,f}} = \rho \left(\frac{\sigma_f - 1}{\sigma_f} \right) \frac{1}{1 + \psi_{i,f}^L} \quad , \quad \frac{w_{i,\text{inf}} L_{i,\text{inf}}}{p_{i,\text{inf}} q_{i,\text{inf}}} = \left(\frac{\sigma_{\text{inf}} - 1}{\sigma_{\text{inf}}} \right) \frac{1}{1 + \psi_{i,\text{inf}}^L}$$

$$\text{and} \quad \frac{r_{i,f} K_i}{p_{i,f} q_{i,f}} = (1 - \rho) \left(\frac{\sigma_f - 1}{\sigma_f} \right) \frac{1}{1 + \psi_{i,f}^K} .$$

We first use data from the 2006 Economic Census to calculate the left hand side of these expressions separately for each firm. Then, we aggregate this expression to the district-sector level, taking the mean of the wage and capital bills across firms in each district-sector.[33]

We estimate separate capital input wedges for the formal sectors in KAPET and non-KAPET districts. Additionally, we estimate labor wedges separately for the formal and informal sectors, assuming that labor production wedges are constant within each sector across locations. Appendix Table A.12 shows that formal capital wedges are lower in KAPET districts than they are in other districts, consistent with the reduced form tax evidence presented in Section 4. It also shows that labor input wedges are substantially smaller for informal firms.

Local Tax Rates. A major component of the formal input wedges are local tax rates, denoted by t_i^L and t_i^K. According to official budget figures, in 2009, Indonesia's total national revenue (from the state budget) totaled Rp 985,725.3 billion. About 70% of national revenues came from domestic taxes, while roughly 26% of national revenues came from non-tax revenue sources, the largest of which consists of natural resource receipts. Our model uses a single local tax parameter, $t_i \equiv t_i^L = t_i^K$, which we assume is constant across inputs, to account for revenue from all of these different sources.

To calculate local tax rates for each district in Indonesia, we first use Ministry of Finance and INDO-DAPOER data to measure each district's own-source revenue. We add to this a measure of each district's locally generated, national tax revenue, and we divide that by total district GDP:

$$t_i = \frac{\text{own-source revenue}_i + \text{locally generated national tax revenue}_i}{GDP_i} .$$

To calculate own-source revenue, we simply use the district's total collected revenue from local taxes (*Pajak Daerah*). However, provinces also collect local taxes, so we distribute provincial own-source revenue to districts using weights equal to the share of each district in provincial GDP.

[31] Note that because of the transfer system for formal tax revenue, aggregate expenditure is no longer equal to aggregate income in each location.

[32] Both values of the elasticity of substitution across varieties are comparable to similar estimates reported by Feenstra et al. (2018) and Simonovska and Waugh (2014).

[33] Note that for the model, we work with a slightly different spatial unit of analysis than we use in the reduced form analysis. Instead of working with coarser 1990 districts definitions, we used district borders in 1998. However, because our model assumes that workers live and work in the same location, we aggregated some large multi-district metro areas into a single location, using urban area definitions described in Civelli et al. (2023). The analysis sample contains 233 districts.

Measuring locally generated tax receipts that contribute to national tax revenue is more difficult. Nationally, we calculate that total tax revenue, after subtracting each district's local taxes, is equal to approximately 7.1 percent of total GDP. After retaining own-source revenues, we assume that each district contributes to national tax revenues in fixed proportion to that district's total GDP, using this national 7.1 percent figure. Districts with higher rates of informality will have lower formal GDP totals, and will contribute less to national tax revenues as a consequence.[34]

Appendix Figure A.5 presents a histogram of t_i across districts. The average formal tax rate is 9.4 percent, and tax rates range from 7.6 to 15 percent. Although poorer districts tend to have higher tax rates than richer districts, richer districts contribute more to overall national tax revenue because they produce higher levels of output.[35]

Regional Public Transfer Rates. To measure net transfer rates, θ_i, we first calculate total district revenues from various fiscal transfer schemes, using data from the Ministry of Finance and INDO-DAPOER. Such schemes include (1) general allocation grants (*Dana Alokasi Umum*); (2) special allocation fund grants (*Dana Alokasi Khusus*), (3) natural resource revenue sharing (*Dana Bagi Hasil Sumber Daya Alam*); and (4) total other revenues. We then choose θ_i to match the following moment:

$$(t_i + \theta_i) \times \text{GDP}_i = \text{own source revenue}_i + \text{fiscal transfers}_i + \text{other income}_i \ .$$

Districts have an average transfer rate of 11.5 percent, and most districts have postive θ_i's, indicating that they are net recipients of transfers. However, 38 districts (16 percent) have negative θ_i's, which means they are donor districts, sending more tax revenue than they spend to other districts through the fiscal transfer system.

Appendix Figure A.6 presents a scatterplot of θ_i against log GDP across districts. This figure shows that wealthier districts have lower θ_i's, while poorer districts have higher θ_i's, as expected.[36] Given that the program targeted lagging regions, it is not surprising that all KAPET districts have positive θ_i's, and that KAPET districts have higher average θ_i's (14.5 percent) than non-KAPET districts (11.3 percent).

Fixed Entry Costs. We use Eq. (15) to solve for fixed costs as a function of the number of firms in location i and sector z (M_{iz}) and total employment by sector (L_{iz}). We calculate both M_{iz} and L_{iz} using 2006 Economic Census data. Appendix Figure A.7 plots kernel density estimates of the distribution of log F_{iz} across districts. Entry costs are significantly larger in the formal sector than they are in the informal sector, as expected.

Appendix Figure A.8 displays the spatial variation in estimates of F_{iz} across districts. Formal entry costs tend to be higher in bigger cities like Jakarta, Surabaya, Medan, and Bandung, and they are lower in East Java and certain parts of the Outer Islands. Informal entry barriers, which exhibit less dispersion, tend to be higher in West Java and Sumatra, but lower in East Java and many parts of the Outer Islands.

Table 6
Migration gravity regressions.

	(1)	(2)	(3)
Log Distance	−1.494***	−1.487***	−1.086***
	(0.020)	(0.033)	(0.054)
Agrosimilarity		0.020*	0.022**
		(0.011)	(0.011)
Absolute Difference in Longitude		0.016	−0.027**
		(0.011)	(0.012)
Absolute Difference in Latitude		0.108***	0.048**
		(0.021)	(0.021)
Cross Island (0 1)		−0.415***	−0.570***
		(0.112)	(0.109)
Same District (0 1)			1.844***
			(0.225)
N of District Pairs	60,516	60,516	60,516
N Clusters	246	246	246
Pseudo R^2	0.799	0.799	0.800
Origin Kabu FE	Yes	Yes	Yes
Destination Kabu FE	Yes	Yes	Yes

Notes: This table shows the results of estimating the gravity equation for migration flows, equation (23). To estimate this equation, we use Poisson Pseudo Maximum Likelihood (PPML) with origin and destination fixed effects, following Correia et al. (2020). Robust standard errors, two-way clustered by origin and destination district, are reported in parentheses. */**/*** denotes significant at 10%/5%/1% levels.

Migration Costs. We parameterize migration costs from location i to location j as follows:

$$D_{ij} = \exp\left\{\beta_d \ln \text{dist}_{ij} + \mathbf{x}'_{ij}\theta\right\} \ , \tag{22}$$

where dist_{ij} measures the physical distance between the centroids of districts i and j and $\mathbf{x}_{ij}$ is a vector of pairwise characteristics, including an indicator for whether i and j are on different islands, the absolute difference in latitude, the absolute difference in longitude, agroclimatic similarity between i and j from Bazzi et al. (2016), and an indicator for whether the origin and destination districts are the same.

Our model admits a gravity equation for migration flows. In particular, the migration probability from location i to j, π_{ij}, in Eq. (4), can be expressed as follows:

$$\pi_{ij} = \exp\left\{\underbrace{\ln\left(\frac{\eta w_j}{\mathbf{P}_j}\right) + \ln(\eta \bar{u}_j)}_{\text{destination FE}} + \underbrace{\ln \sum_{j'} \left(\frac{\bar{u}_{j'} w_{j'}}{\mathbf{P}_{j'}}\right)^{\eta}}_{\text{origin FE}} - \eta\beta_d \ln \text{dist}_{ij} - \eta\left(\mathbf{x}'_{ij}\theta\right)\right\} . \tag{23}$$

To estimate this equation, we use Poisson Pseudo Maximum Likelihood (PPML) with fixed effects, following Correia et al. (2020). Table 6 shows the results, with two-way clustered standard errors by origin and destination district, reported in parentheses.

Column 1 reports a distance elasticity ($-\eta\beta_d$) of −1.5, somewhat larger than other estimates in the literature.[37] We add the pairwise characteristic controls, $\mathbf{x}_{ij}$, in column 2. Interestingly, agroclimatic similarity increases migration flows, even conditional on measures of

[34] In practice, districts contribute differently to national tax revenues because different sectors are taxed at different rates. Our assumption of a constant factor here ignores those distinctions, but is nevertheless consistent with our model.

[35] When we calibrate tax rates in the model, we are using a concept akin to an "effective tax rate", which measures total tax revenues divided by GDP, as opposed to official tax rates which not all firms pay due to informality or special treatment (Besley and Persson, 2014). According to World Bank data, from 2000–2010, Indonesia's total national tax revenue as a share of GDP was 11.86 percent, comparable to our average t_i across districts. Over the same period, India's tax revenue share was 9.9 percent, while China's tax revenue share was 9.7. These similar levels of effective tax rates suggest that the relative lack of impact of Indonesia's KAPET program was not because the country cut taxes when they were already quite small.

[36] Note that by this formulation, we should have that the sum of total transfers across the economy, namely $\sum_i \theta_i \text{GDP}_i$, should be equal to zero. Although we do not impose this condition in calibrating θ_i, our measure of that sum is very close to zero (−0.0017).

[37] For instance, Bryan and Morten (2019) estimate a migration distance elasticity of −0.7, but they pool the 1995 *Supas* (Intercensal Population Survey) together with the 2011 and 2012 *Susenas* for their migration analysis. These are population samples and less comprehensive than full count census data we use.

Table 7
Inverted and observed local fundamentals.

	Productivity		Amenities		Public Goods	Capital
	$\overline{A}_f$	$\overline{A}_{\text{inf}}$	$\overline{u}_f$	$\overline{u}_{\text{inf}}$	G	K
Panel A: Exogenous Fundamentals	(1)	(2)	(3)	(4)	(5)	(6)
Log Distance to Coast	−0.154***	−0.028	−0.422***	−0.171**	−0.301***	−0.327***
	(0.055)	(0.053)	(0.090)	(0.070)	(0.072)	(0.072)
Log Distance to Road	−0.164***	−0.006	−0.144	0.085	−0.596***	−0.591***
	(0.057)	(0.049)	(0.111)	(0.083)	(0.083)	(0.084)
Minning GDP Share	−0.017	0.129***	−0.097**	−0.090*	−0.239***	−0.228***
	(0.047)	(0.049)	(0.048)	(0.051)	(0.055)	(0.054)
Soil bulk density	0.183***	0.103	−0.022	0.078	0.161*	0.147*
	(0.064)	(0.065)	(0.079)	(0.066)	(0.082)	(0.084)
Soil Sand content	−0.153***	−0.062	0.223**	0.255***	−0.133*	−0.138*
	(0.054)	(0.055)	(0.093)	(0.072)	(0.079)	(0.081)
Soil quality: excess salts (HWSD)	−0.123**	−0.049	−0.222**	−0.051	−0.064	−0.094
	(0.048)	(0.047)	(0.099)	(0.077)	(0.068)	(0.068)
Soil quality: toxicity (HWSD)	−0.120**	−0.042	−0.227**	−0.060	−0.061	−0.091
	(0.048)	(0.048)	(0.098)	(0.077)	(0.068)	(0.069)
	A_f	A_{inf}	u_f	u_{inf}	G	K
Panel B: Endogeneous Fundamentals	(1)	(2)	(3)	(4)	(5)	(6)
Electricity Coverage (PLN)	0.204	0.223*	0.016	−0.280**	0.418**	0.428**
	(0.143)	(0.116)	(0.067)	(0.119)	(0.198)	(0.207)
# of Education Facilities Per Capita	0.088	0.170***	0.380***	−0.158*	0.598***	0.612***
	(0.057)	(0.055)	(0.085)	(0.083)	(0.052)	(0.052)
Avg. Distance to Education Facilities	−0.146*	−0.075	−0.329***	0.042	−0.574***	−0.587***
	(0.080)	(0.078)	(0.076)	(0.075)	(0.127)	(0.124)
# of Health Facilities Per Capita	0.163***	0.165**	0.247***	−0.097	0.442**	0.451**
	(0.050)	(0.068)	(0.084)	(0.115)	(0.177)	(0.184)
Avg. Distance to Health Facilities	−0.238***	−0.246***	−0.190*	0.175*	−0.521***	−0.535***
	(0.057)	(0.057)	(0.101)	(0.093)	(0.132)	(0.136)
# of Retail Facilities Per Capita	0.173***	0.211***	0.391***	−0.167*	0.619***	0.640***
	(0.043)	(0.067)	(0.079)	(0.089)	(0.075)	(0.083)
Avg. Distance to Retail Facilities	−0.171**	−0.201***	−0.257***	0.163*	−0.771***	−0.773***
	(0.068)	(0.059)	(0.097)	(0.091)	(0.058)	(0.060)
Island FE	Yes	Yes	Yes	Yes	Yes	Yes
N	233	233	233	233	233	233

Notes: This table provides coefficients from bivariate regressions of estimated location fundamentals and other recovered variables on district-level observables. Each cell in this table is from a separate cross-sectional regression with island fixed effects. The row header denotes the right-hand-side variable name, and the column header denotes the left-hand-side variable name. We standardize both variables using a z-score transformation before estimating regression coefficients. The values of productivity and amenity fundamentals by sector, as well as the values of local public goods and capital, are obtained from the inversion procedure described in Section 6. Sample sizes vary across outcomes but include as many as 233 districts. Robust standard errors are reported in parentheses. */**/*** denotes significance at 10%/5%/1% levels.

physical distance. In column 3, when we also add an indicator for whether the origin and destination district are the same, the distance elasticity approaches −1, consistent with Allen and Arkolakis (2019). Column 3 is our preferred gravity specification.

Inverting the Model. We infer the effective rental rate of capital in each location as $r_{i,f} = \bar{R}(1 + t^K_{i,f})$. Given the Cobb–Douglas form of the production function, we can further derive the levels of capital stocks at each location. Appendix C explains how we invert the model given capital stocks and recover the exogenous components of amenities and productivity in each location and sector.

Appendix Figure A.9 plots the spatial distribution of the inverted local fundamentals, as well as recovered values of capital and local public goods. This figure shows that major urban centers, as well as oil and gas producing regions, tend to have higher levels of exogenous formal productivity, $\overline{A}_{i,f}$, while rural districts and districts in the Outer Islands tend to have lower levels. Informal productivity fundamentals tend to be more evenly dispersed, with greater levels in the Outer Islands than in Java. Amenity fundamentals for both sectors tend to be larger in the Inner Islands, consistent with these locations being important population centers. We also find higher levels of capital and local public goods in Java and Bali, as expected given their concentrations of economic activity.

To confirm the validity of the local fundamentals that we recovered, we correlate them with observed variables that reflect local productivity and amenities from external data. In particular, we use survey data from the 2011 Village Potential Survey (*Podes*), to construct several proxies for amenities and productivity, including distance to retail, education, and health facilities, as well as the share of households connected to the national electricity grid (*Perusahaan Listrik Negara*, or PLN).

Table 7 reports coefficients from bivariate regressions of estimated local fundamentals on district-level observables. Panel A begins by showing that locations farther from the coast have lower levels of formal productivity, lower formal and informal amenities, and lower equilibrium levels of local public goods and capital. The correlations between formal fundamentals and distance to the coast tend to be stronger than correlations between informal fundamentals. We see similar relationships between model recovered attributes and distance to Indonesia's historical network of highways, using data from Gertler et al. (2024), though the latter slopes are not always statistically significant. We also see that informal productivity is higher in areas endowed with more mining resources (as measured by the share of mining in GDP), but mining also seems to reduce amenities, local public goods, and capital.

The next set of rows in Panel A investigate the relationship between soil characteristics and location attributes. Districts with denser and less sandy soils tend to have higher formal productivity, greater local public goods, and higher capital. We also find that soils with excess salinity and higher levels of toxicity are significantly negatively correlated with formal productivity and formal amenities. These soil attributes are also negatively related to informal productivity, informal amenities, local public goods, and capital, though the effects are not significant.[38]

Panel B relates endogenous local fundamentals, local public goods, and capital to measures of government provided infrastructure (electricity, education, and health), as well as retail establishment activity. We find that endogenous productivity is positively related to greater coverage by the national electricity grid, and we find that model implied levels of local public goods and capital are also positively associated with electricity provision. Similarly, the number of education and health facilities per capita are positively associated with formal productivity, formal amenities, local public goods, and capital. Greater average distances to education and health facilities are associated with lower values of these local fundamentals. We also see similar relationships for retail establishments.

Given that the local tax cuts might be expected to encourage local reallocations of economic activity, Appendix Table A.13 compares the recovered local fundamentals between KAPET districts and nearby non-KAPET districts in the same province. This table shows that selected KAPET districts do not appear to be more productive or desirable than nearby Outer Island districts. If anything, KAPET districts tend to have slightly lower levels of formal and informal productivity, but slightly larger levels of amenity fundamentals than nearby non-KAPET districts. This suggests that even if policies to incentivize relocation to KAPET areas may not substantially increase output, they will not necessarily reduce welfare by encouraging people to move to unattractive locations.

7. Counterfactual results

Comparing the Model's ATT to the Reduced Form ATT. Before we analyze the welfare effects of the KAPET program, we first verify that the model's predictions are similar to the reduced form results. To do so, we use our model to simulate a counterfactual scenario of removing the KAPET tax incentives, assuming that the implemented cut to formal capital taxes, t_i^K, was 50 percent in treated areas. Next, we calculate a model-based ATT by taking the average change in outcomes for treated districts and subtracting the average change for non-treated districts in the Outer Islands.[39] We compare both the reduced form and model-based ATTs on five key dimensions: (1) total population; (2) formal employment; (3) informal employment; (4) the number of firms; and (5) district GDP.

Fig. 4 compares the confidence intervals from our preferred reduced form specification (Table 3, Column 4) with the model-based ATT. The dots and bars display the reduced form point estimates and confidence intervals, while the diamonds depict the counterfactual model results. In spite of the sometimes wide confidence intervals, our model's predictions line up quite well with those of the reduced form analysis. Although the reduced form results predict slightly larger effects for formal employment growth, the model-based ATT is still within the 95 percent confidence interval. These results provide a helpful validation check of the model and gives us confidence in using the model to evaluate the KAPET program.

Welfare Impacts of the KAPET Program. We begin by using the same counterfactual exercise to examine how the KAPET program's capital tax cuts impacted overall output growth and welfare. Column 1 of Table 8 shows the welfare and output changes of a 50 percent capital tax break in KAPET districts. We find that the program increased overall welfare by a very small margin (0.01 percent). As a result of the program, KAPET districts experienced a small increase in their total population (0.06 percent), with a 0.62 percent increase in formal sector employment and a 0.50 percent reduction in informal employment. The effect of the KAPET program on total national output was quite small, with an increase of 0.09 percent. While KAPET district GDP grew by 2.3 percent, which is significant, the reallocation of economic activity to these areas resulted in modest GDP losses of 0.02 percent elsewhere.

Appendix Figure A.10 plots spatial variation in different economic outcomes as a result of this 50 percent capital tax cut. Panel A shows that the total population of workers increases in KAPET districts as a result of the tax cut, and this comes from small reductions in nearby districts. Panels B and C show the effects on formal and informal employment. These panels indicate that the program induced in-migration of formal sector employment, mostly from nearby areas, and outmigration of informal employment. Panel D shows that output increases in KAPET districts, with small decreases in output elsewhere.

The literature on place-based policies in developing countries, reviewed by Duranton and Venables (2021), suggests that cuts to factor prices alone may not be sufficient to generate meaningful improvements in local economic activity. In columns 2–4, we evaluate the effects of the KAPET program in a world where productivity and amenity fundamentals in KAPET districts were 20 percent higher, perhaps through local connective infrastructure improvements that facilitate communication or reduce commuting costs. Column 2 of Table 8 shows that by themselves, such infrastructure improvements would have increased KAPET district populations by 2.9 percent, with a 4.2 percent increase in formal employment and a 1.7 percent increase in informal employment. This would generate increases in national output of nearly 1 percent and increase national welfare by 1.5.

Column 3 combines the KAPET tax cuts with the location fundamental improvements and shows that formal population growth in KAPET districts would have increased by 4.8 percent, up from 4.2 percent without the tax cuts. Informal employment growth in KAPET districts would have been smaller, falling from 1.6 percent to 1.2 percent. The combination of policies would have increased national output by 1.1 percent (up from 1 percent without the tax cuts), and welfare would have been improved by 1.6 percent (up from 1.5 percent without the tax cuts). Column 4 shows the difference between columns 2 and 3 and it demonstrates that the KAPET tax cuts program would have been almost 4 times better for national welfare, and 1.3 better for GDP growth, had it been combined with improvements to local

[38] Soil toxicity is measured by the prevalence of calcium carbonate and gypsum. These results on soil characteristics also echo findings from Civelli et al. (2023) that Indonesian cities tended to be settled in places with larger levels of agricultural productivity.

[39] Note that our reduced form estimates report the average annual change in outcomes between treated and non-treated districts. We convert the model based ATT into an annual average ATT assuming the model changes take place over 5 years.

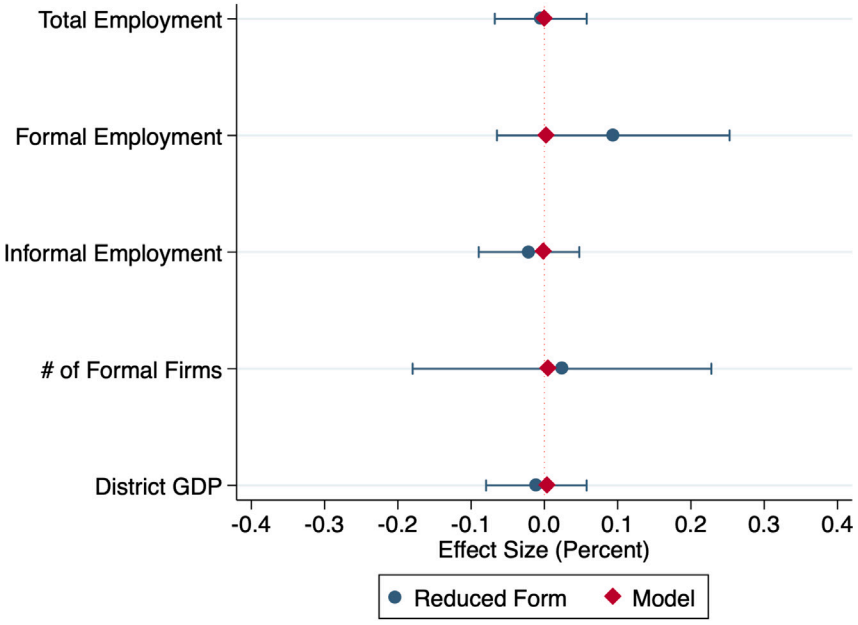


Fig. 4. KAPET program ATTs: Model vs. Reduced form.
Notes: This figure compares confidence intervals from our preferred reduced form specification (Table 3, Column 4) with our model's predicted average treatment effects of the KAPET program on treated districts. The dots indicate the reduced form point estimates, and the bars depict 95 percent confidence intervals. The diamonds are calculated by taking the average change in outcomes for treated districts and subtracting the average change for non-treated districts in the Outer Islands. We convert the model based ATT into an annual average ATT assuming the model changes take place over 5 years. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 8
Evaluating the KAPET program: welfare and GDP changes.

	Baseline	... + 20% Productivity & Amenities			... + 20% Productivity & Amenities ... −20% Migration & Trade Costs		
Panel A: National Outcomes	(1)	w/o KAPET (2)	+ KAPET (3)	net KAPET (4)	w/o KAPET (5)	+ KAPET (6)	net KAPET (7)
... Welfare Change (% Δ)	0.010	1.540	1.579	0.038	2.302	2.343	0.041
... GDP Change (% Δ)	0.093	0.987	1.108	0.121	1.351	1.478	0.127
... Formal Population Share (% Δ)	0.013	−0.047	−0.039	0.008	−0.009	−0.001	0.008
... Formal GDP (% Δ)	0.108	1.026	1.166	0.140	1.436	1.584	0.147
... Informal GDP (% Δ)	0.076	0.942	1.040	0.099	1.252	1.355	0.103
... Capital (% Δ)	0.346	0.001	0.423	0.422	0.074	0.511	0.437
... Local Public Goods (% Δ)	−0.010	0.931	0.916	−0.015	1.393	1.381	−0.012
Panel B: Outcomes in KAPET Districts	(1)	(2)	(3)	(4)	(5)	(6)	(7)
... Population in KAPET Districts (% Δ)	0.060	2.941	3.023	0.082	4.036	4.169	0.133
... Formal Employment in KAPET Districts (% Δ)	0.620	4.214	4.866	0.652	5.498	6.244	0.746
... Informal Employment in KAPET Districts (% Δ)	−0.507	1.651	1.156	−0.495	2.556	2.067	−0.489
... Capital in KAPET Districts (% Δ)	7.229	26.400	35.563	9.164	30.254	39.859	9.605
... Local Public Goods in KAPET Districts (% Δ)	0.788	30.219	31.322	1.103	35.273	36.577	1.304
Panel C: Outcomes in Non-KAPET Districts	(1)	(2)	(3)	(4)	(5)	(6)	(7)
... Population in non-KAPET Districts (% Δ)	−0.003	−0.135	−0.139	−0.004	−0.185	−0.191	−0.006
... Formal Employment in non-KAPET Districts (% Δ)	−0.021	−0.283	−0.311	−0.027	−0.315	−0.347	−0.033
... Informal Employment in non-KAPET Districts (% Δ)	0.010	−0.029	−0.016	0.013	−0.093	−0.080	0.013
... Capital in non-KAPET Districts (% Δ)	−0.080	−1.630	−1.748	−0.118	−1.791	−1.920	−0.130
... Local Public Goods in non-KAPET Districts (% Δ)	−0.056	−0.740	−0.819	−0.079	−0.541	−0.628	−0.087

Notes: Authors' calculations. This table reports percent changes in outcomes, comparing actual values of variables to counterfactual values of variables. Panel A shows percent changes in national outcomes, while Panels B and C focus on percent changes for outcomes in KAPET and non-KAPET districts respectively. In column 1, we compare what would happen if we removed the 50 percent capital tax cut in KAPET districts to actual outcomes. In column 2, we eliminate the KAPET tax cuts but increase location fundamentals for amenities and productivity by 20 percent, and in column 3, we additionally apply the 50 percent tax cut. Column 4 shows the difference in outcomes between columns 2 and 3. In column 5, we eliminate the KAPET tax cuts but increase location fundamentals for amenities and productivity by 20 percent, and we also reduce same-province trade and migration costs by 20 percent for provinces affected by the KAPET program. Column 6 applies the KAPET capital tax cuts under the scenario in column 5, and column 7 reports the difference in outcomes between columns 5 and 6.

fundamentals. These results highlight complementarity between the different policies.

In columns 5–7, we study the KAPET tax cuts combined with both improvements to local fundamentals and also reductions in KAPET province trade and migration costs, as might have occurred if the KAPET program was accompanied by investments in provincial highways. In column 5, we show that the combination of improvements to local fundamentals and reductions in local trade and migration frictions increases national welfare by 2.3 and national GDP by 1.4 percent. Column 6 shows that the capital tax cuts enhance these effects, leading to even larger formal employment growth in KAPET districts (6.2 percent with the cuts vs. 5.5 percent without) and smaller informal employment growth (2.1 percent with the cuts vs. 2.6 percent without). Column 7 shows the difference between columns 5 and 6

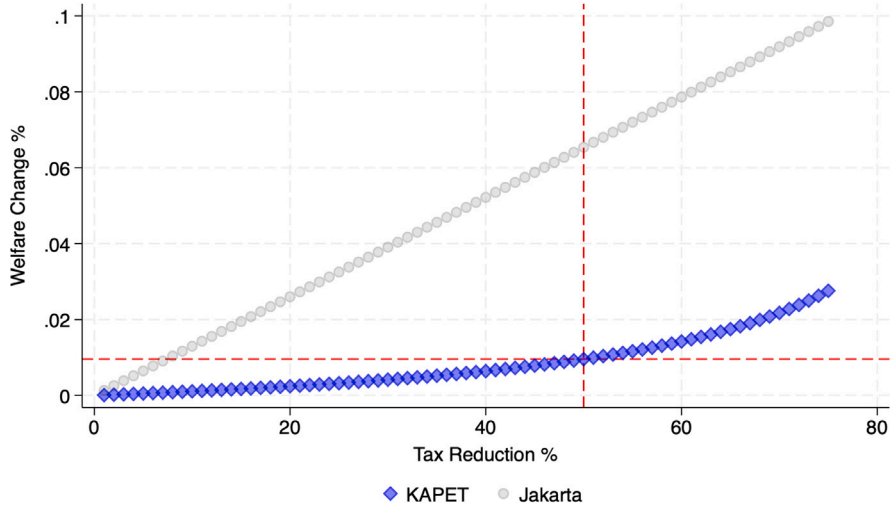


Fig. 5. Welfare changes from tax reductions: KAPET districts vs. Jakarta.
Notes: The blue diamonds in this figure depict the welfare effects of varying sizes of capital tax cuts in KAPET districts. Each black diamond corresponds to the results of a different simulation, where we vary the size of the capital tax cut (plotted on the x-axis). The gray dots depict a different scenario, where we instead apply capital tax cuts to Jakarta. The dashed lines along the y-axis indicate the baseline change in welfare, while the dashed line on the x-axis indicates the baseline KAPET tax cut. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and demonstrates that in a world with better local fundamentals and lower in trade and migration costs, the tax cuts would have been even better for national GDP growth and national welfare. In summary, these counterfactual scenarios suggest that a major reason for the lack of success of the KAPET program was that instead of simultaneously enhancing productivity and amenities or reducing trade and migration costs, the program just cut input costs taxes. Had these cuts been combined with additional improvements, our model suggests that the place-based policy would have been considerably more beneficial for welfare and GDP growth.

Varying the Magnitude of the Tax Cuts. Although we argue in Table 8 that tax cuts alone are unlikely to motivate sufficient reallocation of economic activity, another possible reason why the KAPET program was not impactful is that the capital tax cuts were simply not large enough. In Fig. 5, we explore this hypothesis by plotting national welfare changes across several simulations, where we vary the size of the tax cut that is being applied to KAPET districts (but where we also do not improve local fundamentals or infrastructure). We contrast this set of policies (depicted in black diamonds) with the impact of differently sized tax cuts in Greater Jakarta, the most populous agglomeration in Indonesia (results depicted in gray circles). This figure shows that the welfare effects of KAPET cuts increase with their size, but they are quite small, never exceeding four tenths of one percent even with a 75 percent tax cut. Moreover, the welfare effects of the KAPET tax cuts are dwarfed by similarly sized cuts in Jakarta.[40] This exercise highlights that place-based incentives targeting larger, more productive cities may be more welfare-enhancing than those aimed at underdeveloped areas, like the KAPET program.

Robustness to Varying Parameters. Our previous result suggests that cutting taxes in Jakarta increases welfare by roughly 0.5 percent more than cutting taxes in KAPET districts (i.e. the vertical gap between the Jakarta and KAPET simulations in Fig. 5, Panel A). In Fig. 6, we explore the sensitivity of this welfare gap to different parameter values. We focus on the welfare gap between tax cuts in KAPET districts and tax cuts in Jakarta, because it highlights the different agglomeration and dispersion forces our model emphasizes.

In each panel, every dot corresponds to a different simulation where we rerun both tax cut counterfactuals after varying a single parameter, holding all others constant at their baseline values. We then calculate the impact of the KAPET program on welfare, and to benchmark effects relative to targeting a larger agglomeration, we subtract the welfare impact of the Jakarta tax cut. On the y-axis, we plot this welfare difference; more negative values indicate larger benefits of Jakarta's tax cuts, relative to the benefits of cutting taxes in KAPET areas. The dashed lines intersect the y-axis at the baseline difference in welfare between the two simulations, and they intersect the x-axis at the baseline parameter value.

In panel A, we vary the agglomeration elasticity, α, across a wide range containing many estimates from the literature. Our baseline value of α is 0.056 is similar to Bryan and Morten (2019), but we also explore lower values, such as 0.01 from Rosenthal and Strange (2004), as well as higher values such as 0.076 from Chauvin et al. (2017) and 0.08 from Ahlfeldt and Pietrostefani (2019). This figure shows that as α increases, the welfare benefits of the KAPET program fall relative to cutting taxes in Jakarta. This negative relationship is intuitive, because moving economic activity to KAPET areas encourages relocation away from bigger cities. With stronger agglomeration forces, this relocation becomes more costly to output and welfare. However, the welfare impacts of this change are relatively small.

In panel B, we vary the congestion elasticity, β, across a range that encompasses 0 (Albouy, 2011), −0.04 (our baseline, from Combes and Gobillon, 2015), and −0.075 (Bryan and Morten, 2019). This panel shows that as congestion effects become weaker (and β becomes less negative), the welfare benefits of KAPET relative to cutting taxes in Jakarta also decline. This is also straightforward: one way for KAPET to increase welfare is by moving economic activity out of dense, congested places. If congestion effects are small, the welfare impacts of this kind of reallocation will also be small.

In Panels C and D, we vary the location dispersion parameter, η, and the sector dispersion parameter, ν. These parameters are inversely related to the variance of the Fréchet draws that govern sector and location idiosyncratic heterogeneity. As they decline, workplace and sector choices are governed more by tastes than by changes in location characteristics, such as wages, amenities, or productivity. As expected,

[40] In Appendix Figure A.11, we examine whether the KAPET program would have been more impactful if it had been implemented as policymakers intended. Applying the tax cuts to all 12 KAPET areas from Table 1 (instead of just the "well implemented" districts), we find that welfare impacts of KAPET increase, but they are still less than 2 tenths of 1 percent, even for a 75 percent tax cut.

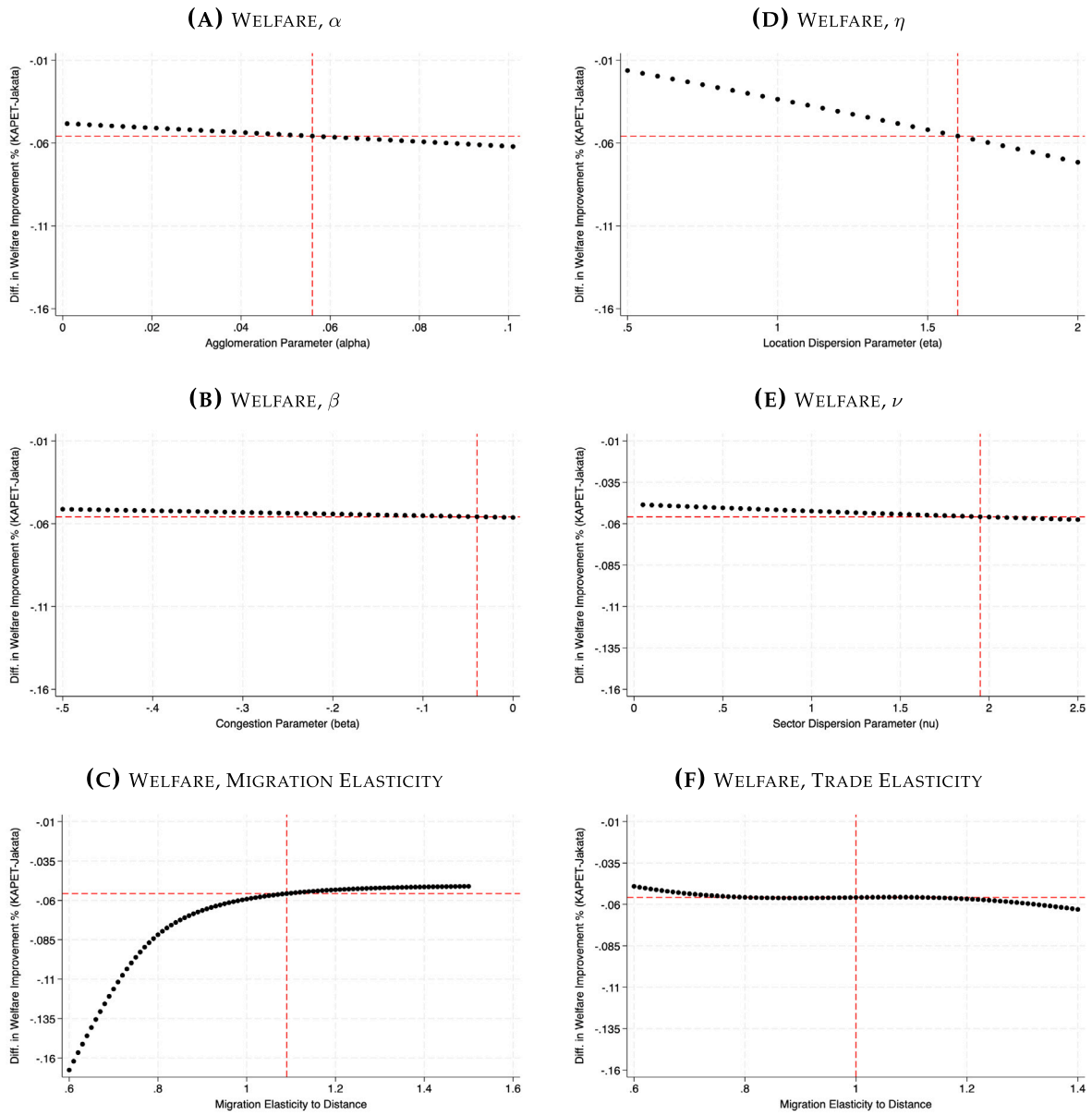


Fig. 6. Welfare impacts of KAPET: Robustness to different parameters.
Notes: This figure explores how different parameter values affect our main conclusions about the welfare effects of the KAPET program relative to capital tax cuts in Jakarta. In each panel, each dot corresponds to the results of different simulations, where we rerun both the KAPET and Jakarta tax cuts but vary a single parameter, holding all others constant at their baseline values. The dashed lines along the y-axis of each panel indicate the baseline change in welfare, while the dashed line on the x-axis indicates the baseline parameter value.

we find that the welfare differences between the KAPET and Jakarta tax cuts fall with lower values of these parameters. This is because tax cuts are less likely to induce relocations if location choices are governed by idiosyncratic heterogeneity.

In Panel E, we vary the migration distance elasticity. Our baseline uses a distance elasticity of −1.086 (from Table 6), but we also explore lower values, such as −0.7 from Bryan and Morten (2019). We find that as this elasticity falls, the welfare gap between KAPET and Jakarta tax cuts grows larger (more negative). This is because when migration costs are less sensitive to distance, tax cuts encourage greater reallocation of workers. In the case of KAPET cuts, this promotes moving people away from places with more agglomeration. Panel F shows that the welfare gap between KAPET and Jakarta tax cuts is fairly insensitive to changes in the trade elasticity, but as trade costs become less sensitive to distance, greater specialization and trade can offset the costs of cutting taxes in KAPET areas relative to Jakarta, and the welfare gap grows smaller. Overall, setting these parameters to different values tends to have small effects, and the results in Fig. 6 suggest that our conclusions about the welfare gap are robust to different parameter values.

Robustness to Alternative Modeling Assumptions. Our model assumes that identical workers make location and sector choices in response to differences in local fundamentals and idiosyncratic preferences. In Appendix Table A.14, we present the results of capital tax cuts in a model with heterogenous workers who differ by skill. In that model, a single sector uses a CES combination of workers and capital to produce output. Details of this model, as well as how we calibrate and estimate parameters, can be found in Appendix E. Overall, we find similar modest welfare and output responses to the capital tax cuts associated with the KAPET program. We also find that the tax

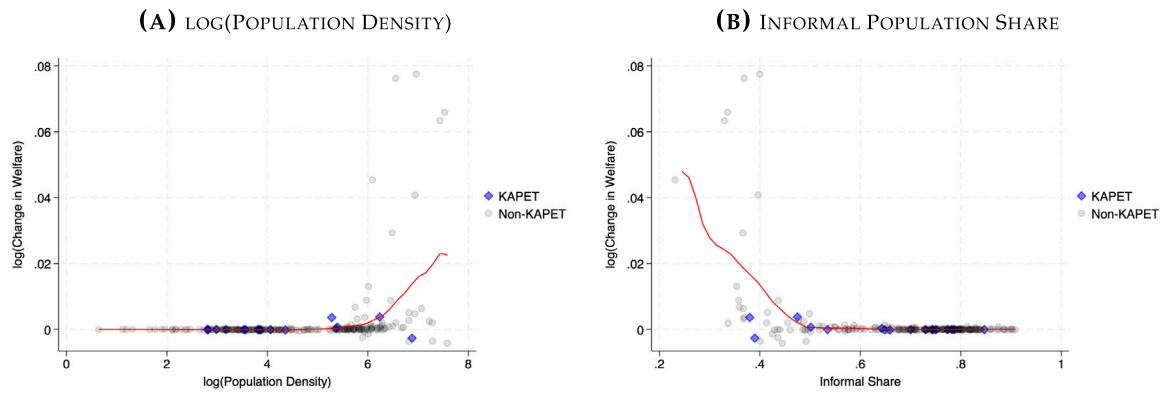


Fig. 7. Welfare effects of single district capital tax Cuts vs. Local characteristics.
Notes: This figure plots the welfare effects of single district tax cuts (along the y-axis) against different values of location characteristics (depicted along the x-axis). Each dot on each scatterplot is from a separate counterfactual where we cut that district's capital taxes by 50 percent. Panel A plots the welfare change of these tax cuts against that district's initial population density, while Panel B plots the welfare change against the share of the population working in the informal sector. The line plots a local linear regression of the welfare effects of single district tax cuts on different values of location characteristics. For this regression, we use an Epanechnikov kernel and a rule-of-thumb bandwidth.

cuts modestly reduce welfare inequality between skilled and unskilled workers.[41]

Evaluating Other Place-Based Policies. Given that the welfare effects of the KAPET program were probably very small and that tax cuts in Jakarta would have been more beneficial, a natural question is whether other place-based policies could have encouraged more productive reallocations of economic activity in Indonesia. To investigate this, we first use our baseline model to conduct counterfactual simulations, applying separate 50% capital tax cuts to every district in Indonesia, one at a time.

Appendix Table A.16 lists the top 10 districts with the largest welfare benefits and the bottom 10 districts that would be the most costly to subsidize. This table tends to show larger welfare benefits for tax subsidies in the biggest cities, like Jakarta, Sarbagita (the Greater Metropolitan area of Denpasar, Bali), as well as trading centers like Batam. This is consistent with the notion that a large share of Indonesia's population is located outside of urban areas and could benefit from more agglomeration. However, we also find significant welfare benefits for some smaller to medium sized cities, like Kupang (Lesser Sunda Islands, population ≈336,000 in 2010). Fig. 7 plots the relationship between welfare changes by district and baseline district characteristics. We see that the welfare effects tend to be larger for tax cuts in denser districts and for tax cuts in districts with smaller shares of informal employment.[42]

Finally, in Table 9, we use our model to explore the welfare effects of other policies targeting KAPET districts. Column 1 replicates our baseline welfare effects of a 50 percent tax cut in KAPET districts (from Table 8, column 1). In column 2, we instead reduce labor taxes by 50 percent in KAPET areas. While this leads to a greater increase in output, it also encourages more migration into KAPET areas, potentially away from other higher amenity places, leading to negligible welfare effects. Column 3 applies a 50 percent cut to both labor and tax cuts. This policy this leads to larger increases in GDP that offset some of the negative welfare effects of the pure labor cuts, but overall welfare effects are similar to column 1.

In column 4, we reduce overall production wedges for both sectors, ψ_{iz}, in KAPET districts by 50 percent. While this leads to even larger increases in output and even greater reallocation of workers, these effects come at the cost of lower national tax revenues. This loss causes large reductions in local public goods and overall welfare losses, despite substantial increases to national GDP. In column 5, we reduce the formal fixed costs that firms pay to enter the market in KAPET districts by 50 percent. This leads to substantial increases in output and welfare, but these gains do not come at the expense of local public goods or national tax revenue. This suggests that efforts to encourage greater entry of firms, either through reducing red tape for starting businesses or by making the formal sector more competitive, would lead to welfare enhancements that do not come at the expense of fiscal pressures.

In column 6, we increase fiscal transfers into KAPET districts by adding 0.05 to each district's net tranfer rate, θ_i. This policy improves local public goods in KAPET districts, leading to an in-migration of workers and large formal output increases. However, local public goods decline nationally, and as a result, national welfare suffers, echoing findings from Henkel et al. (2021) on the effects of Germany's fiscal transfer system. While Indonesia has a long history of using its transfer system to promote regional development (Lewis, 2001), making the system more progressive will restrict public goods provision elsewhere and may not enhance welfare.

8. Conclusion

This paper begins by presenting reduced form estimates of the impact of Indonesia's KAPET program on demographic, employment, and expenditure outcomes, measures of regional output and poverty, and the performance of firms. We find that the program reduced the tax burdens faced by formal manufacturers, but these tax reductions neither increased the entry of large, productive firms nor improved productivity or value added. Consequently, we do not see sustained increases in migration rates or population growth, and overall, treated districts did not grow faster than similar non-treated districts after the introduction of the program. We conduct a synthetic power analysis to demonstrate that our null results are fairly precise. For many outcomes we study, we have the power to reject many effect sizes for similar place-based policies reported in the literature.

To better understand whether the KAPET program could have been designed differently to increase welfare, we calibrate a model which adds capital, sector choices, misallocation wedges, and a fiscal transfer system to a quantitative economic geography framework with trade and migration frictions. After calibrating the model, we conduct a number

[41] Appendix Table A.15 also examines how different modeling assumptions about the transfer system impact our conclusions about the welfare effects of the KAPET program.

[42] Appendix Table A.17 shows bivarate regression results of each district's capital tax welfare change against district characteristics. In addition to relationships described in Fig. 7, we also find larger welfare effects for cutting taxes in districts with: (1) greater levels of formal amenities; (2) greater levels of formal productivity; and (3) smaller values of θ_i.

Table 9
Welfare effects of other policy alternatives.

	Baseline	−50% Labor Tax	−50% Both Labor & Capital Taxes	−50% Prod. Wedges	−50% Formal Fixed Costs	+ 0.05 Transfer Rates
Panel A: National Outcomes	(1)	(2)	(3)	(4)	(5)	(6)
... Welfare Change (% Δ)	0.010	−0.002	0.008	−0.012	0.222	−0.110
... GDP Change (% Δ)	0.093	0.081	0.178	0.912	0.666	0.002
... Formal GDP (% Δ)	0.108	0.170	0.282	1.476	0.943	−0.000
... Informal GDP (% Δ)	0.076	−0.021	0.056	0.254	0.344	0.003
Panel B: Outcomes in KAPET Districts	(1)	(2)	(3)	(4)	(5)	(6)
... Population in KAPET Districts (% Δ)	0.060	0.086	0.147	0.609	0.933	0.117
... Formal Employment in KAPET Districts (% Δ)	0.620	1.911	2.535	11.776	8.157	0.488
... Informal Employment in KAPET Districts (% Δ)	−0.507	−1.763	−2.273	−10.706	−6.387	−0.260
... Local Public Goods in KAPET Districts (% Δ)	0.788	−1.689	−0.945	−9.646	24.579	7.218
Panel C: Outcomes in Non-KAPET Districts	(1)	(2)	(3)	(4)	(5)	(6)
... Population in non-KAPET Districts (% Δ)	−0.003	−0.004	−0.007	−0.028	−0.043	−0.005
... Formal Employment in non-KAPET Districts (% Δ)	−0.021	−0.035	−0.056	−0.260	−0.288	−0.032
... Informal Employment in non-KAPET Districts (% Δ)	0.010	0.018	0.029	0.138	0.132	0.014
... Local Public Goods in non-KAPET Districts (% Δ)	−0.056	−0.106	−0.164	−0.733	−0.611	−0.439

Notes: Authors' calculations. This table reports percent changes in outcomes, comparing actual values of variables to counterfactual values of variables. Panel A shows percent changes for national outcomes, while Panels B and C focus on percent changes for outcomes in KAPET and non-KAPET districts respectively. In column 1, we compare what would happen if we removed the 50 percent capital tax cut in KAPET districts to actual outcomes (replicating Column 1, Table 8). In column 2, we eliminate the capital tax cut but instead cut labor taxes by 50 percent in KAPET districts. In column 3, we combine both labor and capital tax cuts. In column 4, we reduce total production wedges, $\psi_{i,f}^{K}$ and ψ_{iz}^{L} for all sectors z, by 50 percent in KAPET districts. In column 5, we reduce formal fixed entry costs, $F_{i,\text{for}}$, by 50 percent in KAPET districts. In column 6, we increase net transfer rates, θ_i, by 0.05 in KAPET districts.

of different counterfactual simulations to generate a rich set of findings for policymakers who design or administer place-based policies.

First, we find that the KAPET program's capital tax cuts only modestly increased output and welfare. However, we also find that the growth and welfare benefits of these tax cuts would have been enhanced considerably if they were combined with infrastructure improvements that increased productivity and amenities and reduced trade and migration costs. This echoes findings from the literature on place-based policies in developing countries that fiscal incentives, by themselves, may be insufficient to stimulate growth and development (Duranton and Venables, 2021; Frick et al., 2019).

Second, we find that instead of targeting poorer areas, capital tax cuts in areas with greater population density and larger amounts of formal sector employment would have boosted welfare considerably. These findings support the view that it may be difficult for tax incentives to compensate firms adequately for locating in less developed regions, and if anything, place-based policies should attempt to stimulate growth in denser places (Gaubert, 2018; Albouy et al., 2016).

Third, we also use our model to evaluate other place-based policies that may have targeted KAPET districts. Cuts to labor taxes or overall misallocation wedges would have generally had negligible impacts on welfare. However, reducing barriers to formal entry, either by promoting competition in the formal sector or by reducing startup requirements, would lead to substantial growth and welfare enhancements. Finally, we find that while Indonesia's transfer system promotes the delivery of local public goods in poorer places, making the system more progressive may not be growth and welfare enhancing, because it constrains the provision of local public goods elsewhere.

It is clear that more theoretical and empirical research on the effectiveness of place-based policies in developing countries is needed, especially to determine whether such policies could be justified on different grounds, such as reducing unemployment or improving equity. If people have comparative advantages in working in certain locations, the benefits of attracting them to poorer areas with tax cuts or infrastructure improvements may be less clear (Bryan and Morten, 2019). Moreover, our model is static, and dynamic frameworks could prove to be useful in understanding whether place-based policies can help select more optimal steady state equilibria (Allen and Donaldson, 2022). We need more evidence to help policymakers better understand when such outcomes might happen, when they may not, and how to optimize place-based policies to stimulate growth and welfare.

CRediT authorship contribution statement

Alexander D. Rothenberg: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yao Wang:** Writing – review & editing, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Amalavoyal Chari:** Writing – review & editing, Supervision, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jdeveco.2025.103503.

Data availability

Data will be made available on request.

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